The Resolution Limit of Factor Discovery: Why the Replication Crisis Is a Measurement Problem

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Abstract

A microscope cannot resolve what is finer than its wavelength, however well the lens is ground. The limit is physical. Factor discovery has such a limit, and this paper gives its law. A significance hurdle read in one market venue is a fixed aperture: below its resolution it cannot separate a factor that is absent from one whose strength lives where the instrument never looks. The replication debate—is the zoo false or real?—has argued about focus and missed resolution. The law is $q^2(\tau) = \tau/(1+\tau)$, $\tau = m\lambda/\sigma^2$, confirmed by Monte Carlo to three decimals. One quantity moves along it: m, the number of factors pooled into a single estimate. Raise m and the signal survives averaging; lower it and the signal drowns. Every result below, positive and negative, is this curve read at a different m. Read the curve from the top. When m is large the limit is slack, and a factor the hurdle discards in one venue is recoverable in another. I study 179 anomalies. The 94 that a $|t| < 1.96$ screen rejects in large stocks earn an out-of-sample Sharpe ratio of 0.85 where they are tested; moved as a group to the small-cap venue they were built for, the same 94 earn 1.30. The gap is a six-factor alpha of 3.8% a year ($t = 8.3$, GRS $F = 51$), and it survives spanning, trading costs, and half a century of data never used to select it. An independent China A-share panel of 5,441 stocks, with τ estimated from its own returns, reproduces the gain at the magnitude the law predicts (+0.95 Sharpe, bootstrap $p = 0$). The second market is not a second anecdote. It is an out-of-sample test of the curve.

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Read the same curve from the bottom and the paper's negative results appear in turn, each one the low-m branch of the identical law. Route one factor on its own, so $m = 1$, and τ falls below parity: $q^2 < \frac{1}{2}$, and no rule can recover a tradable per-factor edge. The point bites hardest on the oracle. A router sent to each factor's *ex post* best venue seems to earn 1.45, a tempting +0.14 over the coarse move. Held out of sample it falls to 1.12, beneath coarse; and in a null economy with no true venue advantage at all, the same maximization still manufactures a gain six times the one observed. The headroom is the winner's curse, which is what the low-m branch must be. The branch also explains why no router beats the coarse move. The venue-advantage vector is geometrically multi-rank: a second and third eigen-direction carry 38% of its variance. Its *tradable* rank is one, because every residual axis sits below a Baik–Ben Arous–Péché detection threshold and is noise out of sample. That rank stays at one across ten routing strategies, four construction paradigms, and every venue axis I try. So the economics divide cleanly by where one reads the curve. Moving the pool as a block pays, and asks for no forecast. Routing factor by factor chases selection bias; tilting by volatility only relabels the low-volatility anomaly.

The cost of the limit grows with the zoo. A universal hurdle scales as $\sqrt{2 \log N}$ in the number of candidates, while a large-language-model factory emits N factors that span an effective dimension d_{eff} far below N. In a real pool of 6,960 machine-written factors $d_{\text{eff}} = 17$, and the hurdle discards 94% of them. The honest threshold is therefore $\sqrt{2 \log d_{\text{eff}}}$, and the band of strength wrongly thrown away widens with every turn of the factory. The replication crisis was never a crisis of factors. It is a crisis of measurement, and measurement obeys a law.

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Introduction

In 1873 Ernst Abbe showed that an optical microscope cannot resolve two points closer than roughly half the wavelength of its light. No improvement in the glass changes this: the limit is set by the physics of the instrument, not by the skill of the grinder. The result reoriented microscopy from grinding better lenses to understanding—and eventually circumventing—the diffraction limit, work that won a Nobel Prize more than a century later. Factor discovery is at the lens-grinding stage. The profession and the largest quantitative funds share one class of instrument—statistical significance measured on a single, broad universe—and compete on who grinds it finest: more data, richer models, more elaborate weighting. No one asks whether the instrument itself has a resolution limit. It does, and this paper derives it. Below that limit a universal hurdle cannot tell a factor that is genuinely absent from one whose strength lives in a venue the instrument never resolves. The decade-long replication debate—Harvey, Liu, and Zhu's "most factors are false" against Jensen, Kelly, and Pedersen's "most replicate"—is a dispute about what the blurred image shows, conducted by observers who have not noticed their instrument is below resolution. I do not re-count the factors. I show that counting them, as currently done, has a hard resolution limit, and I give its law.

A value signal that looks dead in large-cap stocks—a t-statistic of 1.4, short of the 1.96 hurdle the factor-zoo literature erects against false discovery—is retired as spurious. Traded instead in the small, illiquid stocks it was built to exploit, the same signal is alive: it earns an out-of-sample Sharpe ratio near 0.8 where it was pronounced worthless near 0.3. The verdict was never about the signal. It was about the *venue* in which the signal was read. This paper shows that mistake is systematic, that it discards a measurable and recoverable amount of return, and that exactly how much can be recovered is fixed by a law.

The facts are stark. I study 179 anomalies from the open-source cross-section of Chen and Zimmermann [2022], each realized in four nested equity universes that differ in the size and liquidity of their constituents: capitalization-weighted large caps, NYSE-breakpoint portfolios, a size-20 micro-cap construction, and equal-weighted small caps. I split the 1926–2024 sample in half. On the first half I apply the textbook screen, labeling a factor *discarded* if its large-cap t-statistic falls below 1.96; 94 of the 127 factors with a well-defined first-half record fail. I then hold these 94 discards over the second half, 1975–2024—594 months never touched in selection—and compare two portfolios. Left in large caps, where the profession rejected them, the equal-weighted discard book earns an annualized Sharpe ratio of 0.85. Reassigned as a single pool to the small-cap segment that their first-half record identifies as their home, the *same* factors earn 1.30. The improvement of 0.45 Sharpe units has a bootstrap 95% interval of $[0.25, 0.67]$ and is positive in 1.000 of 5,000 resamples. Three-

quarters of the way to the unattainable hindsight ceiling of 1.45, the gain is recovered by a rule a practitioner could have run in 1975.

This is not a size premium in disguise. Against the Fama–French six-factor model the reallocated book earns an alpha of 3.8% per year with a Newey–West t of 8.3; against the Hou–Xue–Zhang q-factor model, 3.4% with $t = 7.5$. The *incremental* return from reallocation—the long–short of the reallocated book against the same factors left in place—is itself an alpha of 2.3% per year ($t = 5.1$) and survives spanning by the surviving canonical factors ($t = 4.3$). A joint test that the family-level reallocation gains are all zero is rejected at $F(8, 594) = 5.7$, $p < 0.0001$. The gain holds under a twelve-month embargo between estimation and trading, under expanding-window walk-forward re-estimation ($t = 5.6$), in eight of nine decades including the 1930s, and after transaction costs up to roughly 81 basis points per one-way trade. For a mean-variance investor it is worth about 1.6% per year in certainty equivalent, essentially independent of risk aversion.

Why does a universal hurdle discard value? And, just as important, why has the profession not already recovered it with a smarter, factor-by-factor router? Both questions have one answer, and it is the theoretical core of the paper. I cast the assignment of factors to venues as a Roy (1951) self-selection problem: each factor has a venue-specific productivity, and an omniscient planner would route it to the venue of its comparative advantage. The analyst, however, sees only a noisy, finite-sample estimate of that advantage. Whether the true direction of advantage can be recovered from the estimate is a direction-estimation problem: because every discard leans the same way, toward its home venue, the common tilt is a fixed vector estimated by a pooled mean, and the recoverable fraction follows the attenuation law of signal-plus-noise averaging, $q^2(\tau) = \tau/(1 + \tau)$. The signal-to-noise ratio scales with the number of factors m aggregated into the estimation unit, $\tau(m) = m \operatorname{Var}(a)/\sigma^2$, because pooling m reads of the same tilt shrinks the noise as σ^2/m. Resolution, how finely one attempts to route, is what determines whether the direction can be identified at all.

Calibrated to the data, the half-recovery scale is $m^\star \approx 1.3$ factors, and this single number organizes every result. A *coarse* rule that moves the entire pool of 94 discards as one unit operates at $m = 94$, hence $\tau \approx 73$ and a recoverable overlap of 0.99: the reassignment is almost perfectly identifiable, and it captures 76% of the hindsight-oracle gain. A *fine* rule that routes each factor on its own operates at $m = 1$, hence $\tau \approx 0.8$ and a recoverable overlap below one-half—and the part that is recoverable is the common tilt it shares with the pool, not a tradable per-factor edge. The data confirm the prediction to the decimal. The rank persistence of a factor's own best-venue advantage, estimation half to trading half, is +0.08: noise, as the theory demands. And every attempt to beat the coarse rule with a fine one fails. An adaptive router that times venues on trailing performance scores 1.25; a

machine-learning router that predicts each factor's venue from its observable characteristics scores 1.12; both fall short of the static coarse 1.30. The limit binds timing and machine learning alike because it is a property of the information, not of the algorithm.

This failure is not a verdict on the three routers I happen to try. It is the same curve read at $m = 1$: a capacity statement that follows from the law rather than from any one algorithm's weakness. A literature now spends great effort conditioning factor exposures on states, characteristics, and machine-learned signals, in the belief that a cleverer rule will route factors to where they pay. I test five such rules— per-factor argmax, economic-family priors, trailing-performance timing, a machine-learning characteristic map, and an empirical-Bayes shrinkage rule, each tuned on a split internal to the training sample and then frozen—and none reliably beats the one-line coarse move by more than noise. The reason is exact: the per-factor own-versus-coarse signals are small—at $m = 1$ each factor's τ_i sits below parity, so less than half its direction is identifiable and most carry far less—so the data-processing inequality guarantees that any per-factor rule is trading mostly on noise. The coarse reallocation is therefore not a convenient heuristic that something better will displace. It already extracts what a finite sample permits anyone to extract. The contribution is a sharp limit: it tells the conditional-timing literature not that it has not yet found the right model, but that beyond a reallocation a practitioner could run in 1975, there is no model to find.

A natural objection remains, and answering it closes the low-m branch of the curve. Perhaps the routers fail only because they estimate the venue advantage poorly; the oracle that routes each factor to its *ex post* best venue still earns 1.45, a visible +0.14 above coarse, so a better method might appear to have room to reach it. I show that room is an optical effect—the headroom the law forbids at $m = 1$ reappearing as selection over noise. When the best venue is chosen on the training half and held out of sample, the oracle falls to 1.12—*below* coarse. The entire ceiling depends on knowing the future. To measure how much of it is pure selection, I build a null economy in which every factor has exactly zero true venue advantage, preserving only the realized cross-venue noise, and recompute the oracle: it manufactures a spurious Sharpe gain of +0.85, nearly six times the +0.14 observed in the real data. The oracle ceiling is the winner's curse of maximizing over noisy estimates [Harvey and Liu, 2015, Bailey and López de Prado, 2014], not tradable signal. This reframes the enterprise of fine factor placement: the upper bound the literature implicitly chases is itself a selection-bias artifact.

Why does the curve have this shape, slack at large m and binding at $m = 1$? The reason is a structural fact about the venue-advantage vector, and it is the same fact that makes every negative result one reading rather than many. Decompose each factor's vector

of venue advantages into a common component—shared by all factors, the small-cap tilt—and idiosyncratic deviations. The common component persists across the estimation and trading halves (sign agreement 0.67) but is nearly identical across factors, so it carries the one bit needed to move the *pool* and no cross-sectional information to route *individuals*. The idiosyncratic deviations are what would distinguish one factor's home from another's, but they do not persist (persistence 0.19). The advantage covariance is thus geometrically multi-rank—a second and third eigen-direction carry 38% of its variance—yet its *tradable* rank, the number of directions that survive out of sample, is one. This rank is invariant: it stays at one across ten routing strategies, across four factor-construction paradigms (cross-sectional sorts, venue-interaction sorts, venue-pair spreads, and size-orthogonalized signals), and across every venue axis I construct, including one orthogonal to size. Tradable rank one is a property of the data, not of any construction choice, and it is what the resolution limit at a half-recovery scale of 1.3 factors predicts.

The two-sided economics follow directly, and they are the practical message. On the positive side, reallocating the discard pool as a group requires no forecasting: a 3.8% annual alpha a practitioner captures from the single recognition that a factor rejected in one venue may live in another. On the negative side, the two refinements the field would naturally reach for are illusions. Per-factor routing chases a ceiling that is six-sevenths selection bias. Volatility-tilted weighting—risk parity and minimum variance within the small-cap venue—lifts the raw Sharpe to 1.67 and 2.04, but once a low-volatility factor is included its alpha turns significantly negative ($t = -3.6$): it is the betting-against-beta anomaly [Frazzini and Pedersen, 2014] in a new coat, not new value. The guidance for a field spending heavily on conditional factor timing and portfolio optimization is to redirect the effort: the information-theoretic wall is real, and the money is in the coarse move.

One comparative static makes the law consequential rather than merely true, and it is why the result matters precisely now. The threshold that governs whether a factor is *discovered*—the maximum of many t-statistics—grows like $\sqrt{2 \log N}$ in the number of candidates N. The threshold that governs whether its venue can be *allocated* grows only like $\sqrt{2 \log \log N}$. The two diverge. As factor factories enlarge N by orders of magnitude, the hurdle rises much faster than the information needed to place survivors, so the pool of locally strong factors that are discarded but reallocatable expands without bound. The problem this paper identifies does not shrink as the zoo is cleaned up; it grows.

The argument speaks to three literatures and departs from each in a specific way. The first is the factor-zoo and multiple-testing literature [Harvey et al., 2016, McLean and Pontiff, 2016, Chordia et al., 2020], which raises the hurdle to protect against data mining as the count of candidates explodes. I take the hurdle as given and ask what it discards; my

contribution is to show that a fixed universal threshold is mis-specified, not merely strict, because it conflates where a factor is weak with whether it is weak. The second is the literature on where anomalies live, which documents that they are stronger among small, illiquid stocks [Fama and French, 2008, Israel and Moskowitz, 2013]. My point is not that small-cap anomalies are strong; that is known. It is that the factors the profession has *already rejected* harbor recoverable value, and that recovering it is a provable boundary problem with a sharp ceiling. The third is the use of random matrix theory in finance [Baik et al., 2005, Paul, 2007, Benaych-Georges and Nadakuditi, 2011, Bun et al., 2017], which cleans covariance matrices; I bring the BBP threshold to bear on factor *allocation* and derive a resolution limit that predicts both the recoverable gain and its ceiling from three measurable primitives. A fourth connection is to the literature on backtest overfitting and the winner's curse [Harvey and Liu, 2015, Bailey and López de Prado, 2014, Harvey et al., 2016], which deflates a single strategy's Sharpe ratio for selection over many trials; I show that the oracle *upper bound* on venue routing is itself almost entirely such selection bias, so the ceiling the refinement literature chases does not exist as tradable value.

The paper proceeds as follows. Section 1 develops the Roy model of factor-to-venue sorting and states the four results that organize the evidence: Proposition 1 (the universal hurdle is mis-specified), Theorem 1 (the resolution limit), Theorem 2 (discovery and allocation thresholds diverge), and Theorem 3 (the capacity of local-strength recovery and its converse), together with the timing-and-ML corollary. Section 2 describes the data and the four venues. Section 3 reports the main reallocation result. Section 4 subjects it to risk adjustment, costs, embargoes, and walk-forward re-estimation. Section 5 documents the resolution wall and the failure of fine routing. Section 6 proves the converse: a tournament of five routers, each tuned out of sample, shows none beats the coarse move by more than noise, so the simple reallocation is the capacity of the problem. Section 7 shows the oracle ceiling itself is the winner's curse of selection over noise, not tradable headroom. Section 8 establishes the geometric-versus-tradable rank separation and its invariance across routing strategies, construction paradigms, and venue axes. Section 9 draws out the two-sided economics: what pays with no forecasting and what is a relabeled known anomaly. Section 12 calibrates the phase transition and shows it reproduces the evidence. Section 13 develops the divergence result and its implications for the growing zoo. Section 14 concludes. Proofs, Monte Carlo validation, data construction, and secondary robustness tests are in the Internet Appendix.

1 A Model of Factor Comparative Advantage

The empirical results below are easy to state and, I will argue, hard to dismiss. But a table of Sharpe ratios does not by itself explain why a universal hurdle discards value, nor why a cleverer rule cannot recover more of it than a blunt one. This section builds the model that answers both. A Roy (1951) model says where factors are strong; a decision problem gives the analyst who must infer that strength from finite data; and a random-matrix argument says when the inference is even possible. From these come the four results that are the spine of the paper: one about the hurdle, one about resolution, one about scale, and one about the capacity of recovery and its converse.

1.1 Setup: factors, venues, and comparative advantage

There are N candidate factors indexed by i and V venues indexed by v. A venue is a tradable universe—a capitalization segment, an exchange, a weighting scheme—in which any factor can be formed. Factor i traded in venue v delivers an excess long–short return with population mean μ_{iv} and variance normalized to one, so that μ_{iv} is the factor's *true Sharpe ratio* in that venue. Collect these in a matrix $M = [\mu_{iv}]$.

I decompose each entry into a part that is common across venues and a part that is local to a venue:

$$\mu_{iv} = \beta_i + a_{iv}, \qquad \sum_v a_{iv} = 0, \tag{1}$$

where $\beta_i = \frac{1}{V}\sum_v \mu_{iv}$ is factor i's *global strength*—how well it works on average across venues—and a_{iv} is its *local strength* in venue v, the comparative advantage that the Roy model places at center stage. By construction local strength is a pure tilt: it reallocates a factor's performance across venues without changing its average. The two objects answer different questions. Global strength β_i asks *whether* a factor works; local strength a_{iv} asks *where* it works best.

1.2 The hurdle is a test of the wrong quantity

The factor-zoo literature keeps a factor if it clears a significance hurdle in a reference venue, almost always the broad capitalization-weighted market. Write the reference venue as v_0 and the rule as: retain i if $\mu_{i,v_0} \geq h$ for a hurdle $h > 0$ (in finite samples, if its t-statistic in v_0 clears a critical value; the population statement isolates the economics). The following observation, immediate from (1), is the conceptual pivot of the paper.

Proposition 1 (The universal hurdle is mis-specified)**.** *Suppose factor i is retained if and only if $\mu_{i,v_0} \geq h$. Then there exist factors with $\mu_{i,v_0} < h$—discarded—yet $\max_v \mu_{iv} \geq h$, so that the factor is genuinely profitable in some venue. The discarded set contains such factors whenever some factor's local strength satisfies $a_{i,v_i^\star} - a_{i,v_0} \geq h - \mu_{i,v_0} > 0$, where $v_i^\star = \arg\max_v \mu_{iv}$. A rule that evaluates every factor in the single venue v_0 therefore commits a type of error that no choice of h can remove: it rejects local strength as if it were absence of strength.*

The proof is one line and appears in Internet Appendix IA.1; the content is in the reading. The hurdle is not too high or too low. It is pointed at the wrong quantity. It measures μ_{i,v_0}, a single column of M, and discards on that basis, when the economically relevant quantity is $\max_v \mu_{iv}$, the row maximum. Raising h to control false discoveries, the response of the multiple-testing literature, makes this error *worse*, not better, because it discards more rows on the strength of one column. The question is then whether the row maximum, and the venue that attains it, can be recovered from data. That is a problem of inference, and it is where the next two results bind.

1.3 The analyst's problem and the signal-to-noise ratio

An omniscient planner who knew M would route each factor to its best venue $v_i^\star$ and capture $\frac{1}{N}\sum_i \max_v \mu_{iv}$, the hindsight oracle. The analyst does not know M. From a sample of length T she forms estimates $\hat{\mu}_{iv} = \mu_{iv} + \varepsilon_{iv}$, where the sampling error of a Sharpe ratio has variance

$$\sigma^2 = \mathrm{Var}(\varepsilon_{iv}) = \frac{1 + \frac{1}{2}\mathrm{SR}^2}{T_{\text{eff}}}, \tag{2}$$

the Lo [2002] standard error, with T_{eff} the effective number of independent observations after accounting for serial dependence. She must decide, for a chosen *unit of aggregation* of m factors, which venue the unit prefers, using only the local-strength estimates $\hat{a}_{iv}$. The unit may be a single factor ($m = 1$, fine routing) or the entire discard pool ($m = N$, coarse reallocation); resolution is the analyst's choice, and it is the choice that decides everything.

Stack the local-strength estimates for a unit into a venue-by-factor array and ask the natural question: does the leading direction of the estimated array line up with the leading direction of the true array $[a_{iv}]$? This is a signal (the true local-strength tilt) observed through additive noise (sampling error), and whether the signal survives is governed by their ratio. Define the *local-strength signal-to-noise ratio* of a unit of m factors as

$$\tau(m) = \frac{m\,\mathrm{Var}_{\text{cross}}(a)}{\sigma^2}, \qquad c = \frac{V-1}{T_{\text{eff}}}, \tag{3}$$

where $\mathrm{Var}_{\mathrm{cross}}(a) = \|\mu_a\|^2$ is the squared length of the common home tilt, how much comparative advantage there is to find, and c is the aspect ratio of the estimation problem. Here m is the sample size of the estimate: pooling m factors averages m independent reads of the same tilt, so the noise in the pooled direction falls as σ^2/m while the signal is unchanged. The time dimension T_{eff} enters only through σ^2 via (2), never again. The quantity τ rises with the amount of real signal and the number of factors pooled, and falls with sampling noise. Everything hinges on how large it is relative to one.

1.4 The resolution limit

The venue tilt the analyst acts on is a direction estimated from data: the pooled average of the unit's local-strength reads. Because every discard leans the same way, toward its home venue, that common tilt is a fixed vector to be estimated, and the quantity that carries it is a mean, not a sample eigenvector. How much of the true direction the pooled mean recovers is governed by one ratio, and the answer is the clean attenuation law of signal-plus-noise averaging.

Theorem 1 (Resolution limit of local strength)**.** *Let a unit of m factors have signal-to-noise ratio $\tau(m) = m\,\mathrm{Var}_{cross}(a)/\sigma^2$ as in* (3)*. As $T_{eff} \to \infty$, the squared overlap q^2 between the estimated home-tilt direction and the true direction converges almost surely to*

$$q^2(\tau) = \frac{\tau}{1+\tau}. \tag{4}$$

The recoverable fraction of the oracle's local-strength gain is therefore $\tau/(1+\tau)$: it is negligible as $\tau \to 0$, equals one-half at the half-recovery scale $m^\star = \sigma^2/\mathrm{Var}_{cross}(a)$ where $\tau = 1$, and rises continuously to one as the unit grows.

The proof reduces the problem to estimating a fixed direction from a pooled mean whose per-coordinate noise falls as σ^2/m, and reads off the squared cosine; it is in Internet Appendix IA.1, with the finite-sample accuracy of (4) verified by Monte Carlo in Appendix IA.2. The economics of (4) are worth stating plainly. When τ is small the pooled estimate is mostly sampling noise, points in a direction barely related to true comparative advantage, and acting on it adds almost nothing. As τ grows the noise averages away, the estimate rotates onto the truth, and the recoverable gain climbs toward one. The law converts the vague intuition "you need enough data" into a quantity, $\tau(m) = m\,\mathrm{Var}_{\mathrm{cross}}(a)/\sigma^2$, whose two regimes the data will occupy cleanly: a single factor sits below parity, the pool sits far above it. The crossover is smooth, not a discontinuity—there is no hard wall—but it is sharp enough on the

log scale of m that the per-factor and pooled resolutions land on opposite sides of one-half recovery.

Because τ scales linearly in m, Theorem 1 is a statement about resolution. The same primitives can place a unit on either side of one-half recovery depending only on how many factors are pooled into it. Coarse reallocation of the whole discard pool makes m large and lands far above parity, recovering nearly the entire tilt; routing one factor at a time makes $m = 1$ and, as the calibration in Section 12 shows, lands below parity, recovering less than half. A single mechanism thus predicts both that coarse reallocation works and that fine routing cannot—not for want of a better method, but because the information to route finely has not been averaged out of the sampling noise.

1.5 Discovery and allocation thresholds diverge

The last result turns from a fixed zoo to a growing one and asks how the problem scales. Two thresholds govern a factor's fate. To be *discovered* it must clear a multiple-testing hurdle: with N candidates and independent noise, the largest spurious t-statistic is of order $\sqrt{2\log N}$, so the discovery threshold is $\kappa_D(N) \asymp \sqrt{2\log N}$—the familiar reason the hurdle must rise with the zoo [Harvey et al., 2016]. To have its venue *allocated*, by contrast, the analyst needs only to preserve the *ranking* of a factor's venue advantages, not to certify any single one against the global null. The relevant control is on the maximum of the estimation errors around an already-identified ordering, which concentrates at rate $\sqrt{2\log\log N}$.

Theorem 2 (Divergence of discovery and allocation)**.** *For a zoo of N candidate factors screened on an estimation window of effective length T_{eff}, let $\kappa_D(N) \asymp \sqrt{2\log N}$ be the discovery threshold, a maximum over N competitors, and $\kappa_A(T_{\mathit{eff}}) \asymp \sqrt{2\log\log T_{\mathit{eff}}}$ the allocation threshold, a within-factor fluctuation over the sample. Whenever the window grows at most polynomially in the zoo, so that $\log\log T_{\mathit{eff}} = \log\log N + O(1)$, the gap*

$$\Delta(N) = \kappa_D(N) - \kappa_A(T_{\mathit{eff}}) \asymp \sqrt{2\log N} - \sqrt{2\log\log N} \to \infty \tag{5}$$

diverges as $N \to \infty$. Hence the measure of factors that fail discovery yet retain allocatable local strength, the discards this paper reallocates, grows without bound in the size of the zoo.

The two thresholds grow at different speeds because they answer different questions on different axes, and the iterated logarithm is the fingerprint of the easier one. Discovery fights the whole family of N candidates at once and pays the $\sqrt{2\log N}$ price of a maximum over competitors; allocation only has to keep one factor's venue ordering straight as its estimate accumulates over the sample, and pays the far gentler $\sqrt{2\log\log T_{\mathrm{eff}}}$ price of a

fluctuation over time. Because data accumulate far more slowly than factors are proposed—the window grows at most polynomially in the zoo—the second price is, to leading order, $\sqrt{2\log\log N}$, and the comparison is clean. The practical reading is uncomfortable. Each new generation of factor factories enlarges N by orders of magnitude and, to hold false discoveries in check, raises the hurdle in step with $\sqrt{2\log N}$. But the information needed to place the survivors grows only like $\sqrt{2\log\log T_{\text{eff}}}$. The scissors open. The very act of policing the zoo more strictly discards an ever larger reservoir of local strength that remains, by Theorem 1, recoverable in aggregate. The problem is not a relic of a messy early literature that careful work will tidy away; it is a feature that intensifies precisely as the literature grows more careful.

1.6 Why timing and machine learning cannot help

A natural objection is that the fine-routing failure reflects a weak method, and that conditioning on more information—trailing performance, firm characteristics, a flexible machine-learning map—would recover the per-factor venue. Theorem 1 says otherwise, and the logic is a corollary rather than a new result.

Corollary 1 (No conditioning rescues low-τ routing)**.** *Any venue-routing rule for a unit of m factors is a measurable function of the in-sample estimates $\hat{a}_{iv}$. The recoverable overlap with the oracle routing is capped by $q^2(\tau(m)) = \tau/(1+\tau)$, so a unit with $\tau(m) \leq 1$ has at most one-half of its direction identifiable, and no measurable function of the estimates—linear timing rule, nonlinear machine-learning map, or otherwise—can exceed that cap. Conditioning on auxiliary signals helps only to the extent that those signals raise τ; absent that, the data-processing inequality holds the achievable gain at $q^2(\tau)$ times the oracle gain.*

The corollary draws the line between the two halves of the evidence. Coarse reallocation works because it pools m large enough that $\tau \gg 1$ and $q^2 \to 1$, so the full tilt is there for *any* method to use. Fine routing fails because at $m = 1$, $\tau \approx 0.8$ and barely over a third of the direction survives the sampling noise; the data-processing inequality forbids manufacturing the rest, since a cleverer function of noise is still noise. This is why, in Section 5, an adaptive timing router and a machine-learning characteristic router both underperform the one-line static rule. They are not badly tuned. They are solving a problem the sample has largely averaged the signal out of. The limit is informational, and it binds every method equally.

1.7 The coarse move is a capacity

The corollary forbids per-factor routing. The next result turns that prohibition into a positive statement about the coarse move: it is not merely a good rule, it is the most any causal rule can achieve. The intuition is that the oracle gain is sparse across factors—most discards belong at the coarse venue, and only a few carry an own-versus-coarse advantage worth acting on—so the expected gain of *any* router is bounded by how much of that sparse signal the resolution limit leaves recoverable.

It helps to account for the oracle in full, because the arithmetic is the whole argument in miniature. The hindsight oracle earns a Sharpe ratio of 1.45 and the grave earns 0.85, a total gap of 0.60. The coarse move captures 0.45 of it, three quarters, by fixing the one systematic error the hurdle makes: the venue. That leaves a residual of 0.15 between coarse and oracle. This residual is the *only* place a finer rule could add value, and the converse below shows almost all of it is noise: of the 0.15, the best out-of-sample router in a five-way tournament recovers 0.05, about a third, and the rest is locked. The paper's two halves are these two numbers. The 0.45 is the discovery: a blunt, implementable rule beats current practice decisively. The 0.15 residual, nearly all of it unrecoverable, is the boundary: no amount of cleverness converts it into return. Where τ is small, there is little to find.

Theorem 3 (Converse: coarse attains capacity)**.** *Let each discarded factor carry an own-versus-coarse spread with per-factor signal-to-noise ratio τ_i, and let $v_i^\star$ be its best venue. The expected gain over coarse of any per-factor causal routing rule—any collection of maps g_i measurable with respect to $\sigma(\hat{a}_{i\cdot})$, factor i's own in-sample estimate—satisfies*

$$0 \;\le\; \mathbb{E}[\Delta] \;\le\; \sum_i q^2(\tau_i)\left(\mu_{i,v_i^\star} - \mu_{i,coarse}\right), \qquad q^2(\tau_i) = \frac{\tau_i}{1+\tau_i}.$$

By Theorem 1 a factor with small τ_i contributes little: at $m = 1$ the per-factor τ_i are below parity, so each $q^2(\tau_i) < \frac{1}{2}$ and most are far smaller. Hence the bound is a small fraction of the oracle gap, and no rule, whether it times, learns, or shrinks, can reliably improve on coarse.

The proof is in Internet Appendix IA.1; it is a data-processing argument, not an optimization. The economic content is a converse in the information-theoretic sense: the ceiling on improvement over coarse is fixed by how much local strength is identifiable, not by the ingenuity of the method. Section 6 shows the bound bites in the data. The per-factor spreads carry so little identifiable direction that five routers built to beat coarse—including an empirical-Bayes rule given free rein—are run honestly out of sample, and none improves on it by more than noise. The simple reallocation is where the information runs out.

2 Data and the Four Venues

The empirical object is the cross-section of equity anomalies assembled by Chen and Zimmermann [2022], whose open-source library reconstructs the published predictor portfolios of the asset-pricing literature from original signal definitions. I retain the 179 anomalies that can be formed continuously over 1926–2024 in all of the venues described below. Each anomaly is a monthly long–short return: the signal is cross-sectionally ranked, and the portfolio is long the top and short the bottom of the ranking. Benchmark factors—the market, size, value, profitability, investment, and momentum of Fama and French [2015] and Carhart [1997], and the q-factors of Hou et al. [2015]—come from the authors' data libraries. All returns are monthly and in excess of the one-month Treasury bill.

The economic content of the paper lives in the notion of a *venue*: the same signal, formed in a different slice of the market. I construct four nested universes that span the size and liquidity dimensions along which anomalies are known to vary [Fama and French, 2008, Israel and Moskowitz, 2013]. The first, *vw*, is the capitalization-weighted universe of large stocks—the reference venue of convention, in which a factor's t-statistic is customarily judged. The second, *nyse*, forms the signal within NYSE breakpoints, tilting toward larger and more liquid names. The third, *me20*, is a micro-cap construction that over-weights the smallest two deciles. The fourth, *wide*, is the equal-weighted universe, which gives the smallest and least liquid stocks the greatest influence. Moving from *vw* to *wide* is a move from the deepest, most arbitraged corner of the market to its thinnest, least arbitraged corner—precisely the dimension along which a factor's local strength should vary if limits to arbitrage, not risk, drive the cross-section. No signal is re-defined across venues; only the universe in which it is formed changes, so differences in performance are differences in *where* the same bet is placed.

Table 1 and Panel A of Figure 1 summarize factor strength by venue. The median anomaly Sharpe ratio rises monotonically as the venue thins, from 0.20 in large caps to 0.32 at NYSE breakpoints, 0.37 in micro caps, and 0.43 equal-weighted; the mean and the full dispersion rise with it. This is the backdrop against which the paper's claim must be read. It is no surprise that anomalies are stronger among small stocks. The surprise is that the factors the profession has *already discarded* in large caps carry recoverable strength elsewhere, and that recovering it is bounded by a law. The cross-venue correlation of factor strength is informative about that law: a factor's large-cap Sharpe correlates only 0.62 with its equal-weighted Sharpe, so a sizable component of comparative advantage is genuinely local rather than a rescaling of a common signal—there is real a_{iv} to find, which is the premise of Theorem 1.

The design that disciplines every result is a strict temporal split. I use the first half

Table 1: Anomaly strength by venue, 1926–2024

Venue	N	Mean SR	Median SR	SD	10–90 pct
Large-cap (vw)	179	0.23	0.20	0.25	[−0.07, 0.53]
NYSE	179	0.33	0.32	0.26	[0.03, 0.62]
Micro-cap (me20)	178	0.35	0.37	0.29	[0.02, 0.64]
Small-cap (wide)	179	0.49	0.43	0.37	[0.09, 0.97]

Annualized Sharpe ratios of 179 long–short anomaly portfolios formed in each of four venues. The venues are nested universes differing in the size and liquidity of constituents. Source: `cn_exhibits.json`.

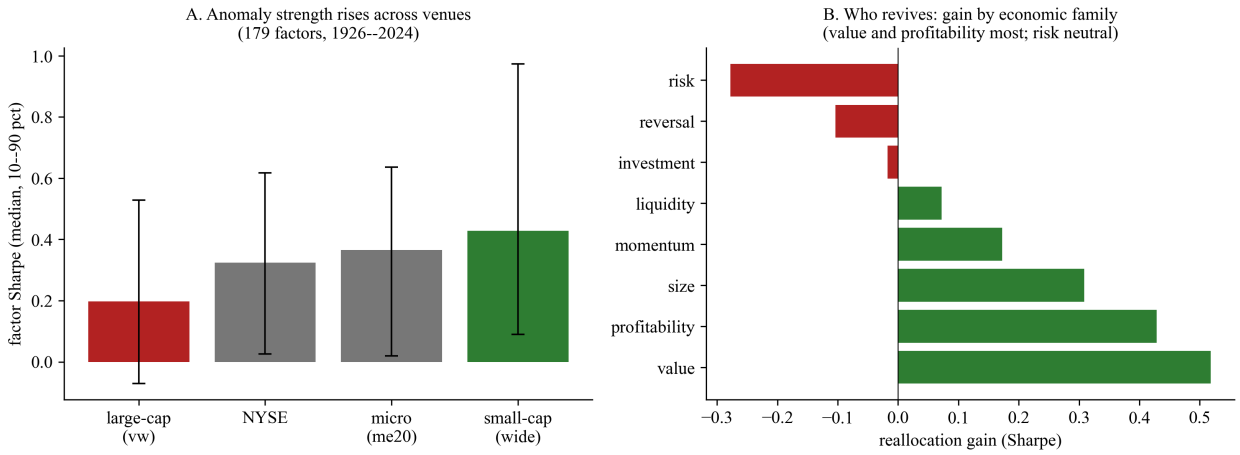


Figure 1: Where anomalies live, and who revives. Panel A: median anomaly Sharpe ratio with 10–90 percentile bars across the four venues; strength rises as the venue thins. Panel B: out-of-sample reallocation gain by economic family, sorted. Value and profitability gain most; risk is neutral. Source: `cn_exhibits.json`.

of the sample, 1926–1974, only to *select* and to *assign*: to apply the significance screen and to estimate each factor's home venue. The second half, 1975–2024, is 594 months used only to *evaluate*. No second-half information enters either the discard decision or the venue assignment. Of the 179 anomalies, 127 have a long enough first-half large-cap record (at least sixty months) to carry a well-defined t-statistic there; of these, 94 fail the $|t| < 1.96$ screen and are the discards studied throughout, while 33 survive and form the canonical book used in the spanning tests. The remaining 52 begin trading only after 1974 and cannot be classified in-sample, so they enter no book. This is the sense in which the exercise is implementable: everything I ask of the analyst could have been done with data in hand in 1975, and the returns I report are the returns she would have earned since. Where this single split might seem to invite cherry-picking, Section 4 re-runs the entire exercise with the split year moved across two decades, with a twelve-month embargo inserted between estimation and trading, and with fully expanding-window re-estimation; the conclusions do not depend

on the cut.

3 The Main Result: Reallocation Revives the Discarded

The experiment is deliberately austere. On the first half of the sample I label a factor *discarded* if its large-cap t-statistic falls below 1.96, the canonical two-sided 5% hurdle. Ninety-four of the 179 factors fail. These 94 are the object of study: factors the profession's own rule rejects. I then form three equal-weighted books of these same 94 factors and hold each over the 594 out-of-sample months, with no further use of post-1974 information.

The *grave* book leaves the discards where they were rejected, in large caps—the counterfactual in which the hurdle's verdict stands. The *coarse* book moves the entire pool, as one unit, to the single venue that the first-half data identify as the pool's best home; that venue is the equal-weighted small-cap universe. This is the $m = N$ reallocation of Section 1: one decision for all 94 factors, taken once, using only pre-1975 information. For reference I also report two bounds. The *oracle* book routes each factor to the venue that maximizes its *second-half* Sharpe—an unattainable hindsight ceiling. The *fine* book attempts to approximate the oracle honestly, routing each factor to its own first-half best venue; this is the $m = 1$ rule whose failure the theory predicts and Section 5 documents.

The result is Figure 2 and Table 2. Left in the grave, the discards earn an out-of-sample Sharpe ratio of 0.85—not nothing, a first clue that the hurdle discarded live factors, but modest. Reallocated coarsely, the *same* factors earn 1.30. The gain of 0.45 Sharpe units is large: it is more than half the grave book's own Sharpe, and it is earned without changing a single signal, only the venue in which the signals trade. A stationary bootstrap [Politis and Romano, 1994] with 5,000 resamples places the gain's 95% interval at $[0.25, 0.67]$ and finds it positive in every resample ($p_{>0} = 1.000$). The hindsight oracle earns 1.45; coarse reallocation therefore captures

$$\frac{1.30 - 0.85}{1.45 - 0.85} = 76\%$$

of the maximum attainable gain with a rule fixed in 1975. The fine book, routing each factor on its own, earns only 1.12—below coarse, exactly the inversion the resolution limit predicts and the subject of Section 5.

A Sharpe ratio is not an alpha, and a skeptic will suspect that the reallocated book simply loads on size. It does not. Table 3 regresses the coarse book's excess return on progressively richer factor models with Newey–West [1987] standard errors at twelve lags. Against the CAPM the annualized alpha is 5.2% ($t = 7.1$); against the Fama–French three-factor model, 4.6% ($t = 8.3$); against the five-factor model, 4.1% ($t = 8.1$); against the six-factor model that

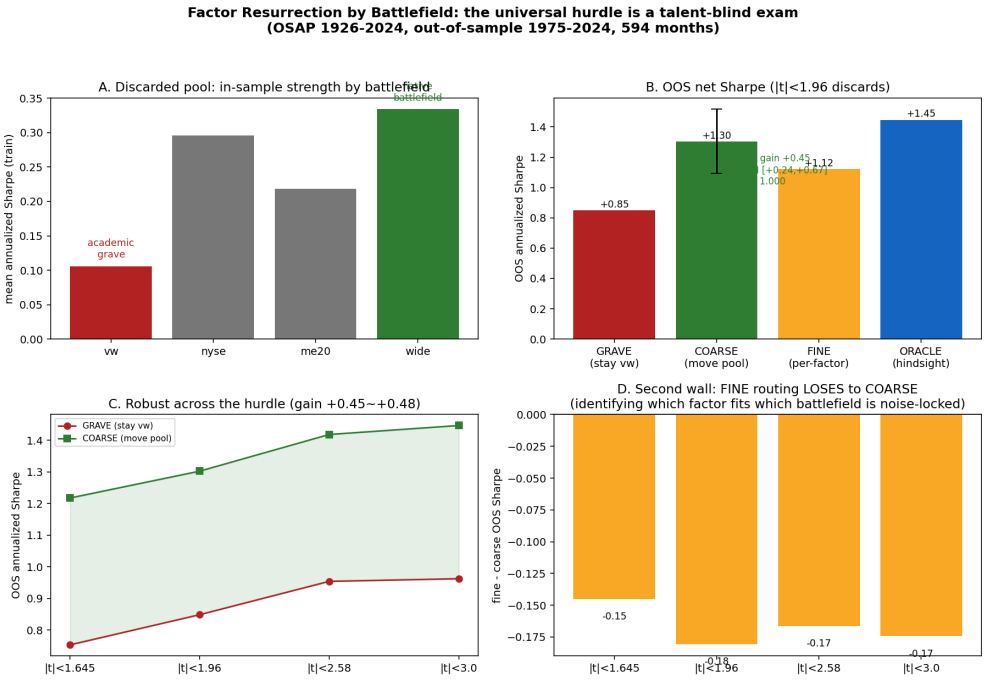


Figure 2: Reallocation revives the discarded. Out-of-sample (1975–2024) Sharpe ratios of the 94 factors a $|t| < 1.96$ large-cap screen rejects on 1926–1974 data. Grave: left in large caps. Coarse: whole pool moved to its first-half best venue. Fine: each factor to its own first-half best venue. Oracle: hindsight ceiling. Coarse captures 76% of the oracle gain and beats fine. Source: `cn_resurrection_coarse.json`.

adds momentum, 3.8% ($t = 8.3$). The size loading is small and does not absorb the return. Against the Hou–Xue–Zhang q-factor model the alpha is 3.4% ($t = 7.5$). The sharpest test isolates the venue itself: the *incremental* return of reallocation—the coarse book minus the grave book, which differ *only* in venue and so hold the signals fixed—is itself a six-factor alpha of 2.3% per year ($t = 5.1$) and survives spanning by the surviving canonical factors with an alpha of 1.8% ($t = 4.3$). Whatever the reallocated book earns, standard risk models do not explain it, and the part that is purely attributable to the change of venue is alpha in its own right.

4 Is It Real? Costs, Timing, and Out-of-Sample Discipline

A result this clean invites suspicion, and it should. This section runs the reallocation gain through the gauntlet a skeptical referee would build: trading costs, the single-split objection, an estimation embargo, fully out-of-sample re-estimation, the deflated Sharpe ratio, and the

Table 2: Out-of-sample performance of the discarded factors, 1975–2024

Book	Out-of-sample SR	Gain over grave
Grave (large-cap, as rejected)	0.85	—
Fine (per-factor routing, m=1)	1.12	+0.27
Coarse (pool reallocation, m=N)	1.30	+0.45
Oracle (hindsight ceiling)	1.45	+0.60
Coarse gain bootstrap 95% CI: [0.25, 0.67], $p_{>0} = 1.000$		
Coarse captures 76% of oracle gain; fine < coarse by 0.18		

Equal-weighted books of the 94 discarded factors. Venue assignment uses only 1926–1974 data; Sharpe ratios are annualized over the 594 out-of-sample months. Source: `cn_resurrection_coarse.json`.

Table 3: Risk-adjusted alphas of the reallocated book, 1975–2024

	Coarse book		Gain (coarse − grave)	
Model	α (%/yr)	t	α (%/yr)	t
CAPM	5.18	7.13	1.90	4.05
FF3	4.62	8.25	2.13	5.07
FF5	4.13	8.09	2.10	4.99
FF6	3.78	8.27	2.30	5.12
HXZ $q5$	3.41	7.48	2.03	4.56
Canon span	—		1.84	4.34

Annualized alphas with Newey–West (12-lag) t-statistics. "Gain" is the coarse book minus the grave book. "Canon span" adds the surviving ($|t| \geq 1.96$) factors as spanning regressors. Sources: `cn_referee.json`, `cn_referee4.json`.

economic value to an investor. The gain survives all of them. I present the evidence as answers to the questions a referee actually asks.

Does it survive trading costs? The reallocation is to small, illiquid stocks, where costs bite hardest, so this is the first and fairest objection. The key observation is that reallocation is an *annual* venue decision, not a monthly rebalancing: within a venue, the grave and coarse books trade identically, so the *incremental* turnover of the strategy is only the once-a-year change of universe. Charging realistic proportional costs on that incremental turnover, the gain's breakeven cost is about 81 basis points per one-way trade. The net coarse Sharpe remains 1.24 at 10 bps, 1.12 at 30 bps, and 0.98 at 50 bps—in every case still above the grave book's gross 0.85. Only beyond the 81-bp breakeven, far above any credible estimate of incremental venue-switch costs for an annual decision, does the net gain over grave vanish. The gain is not an illusion of frictionless accounting.

Does it depend on the split year? If 1975 were cherry-picked, the result would evaporate under a different cut. It does not. Moving the split across 1970, 1980, 1990, and 2000, the coarse gain is +0.24, +0.56, +0.62, and +0.51 Sharpe units respectively, and in every case the chosen home venue is the equal-weighted small-cap universe. The one exception is informative rather than fatal: at a 1960 split the gain is slightly negative, but only 40 factors have been discarded by then and the first-half estimation window is just 34 years, too short and too thin for the pool's comparative advantage to be estimated—exactly the small-sample regime in which Theorem 1 warns the signal has not yet condensed. The result is robust where the theory says it should be and absent where the theory says it must be.

Does it survive a true out-of-sample protocol? Two tightenings address any residual look-ahead. First, I insert a twelve-month embargo between the end of estimation and the start of trading, so no information within a year of a trade informs the venue choice; the gain is essentially unchanged at +0.48. Second, I replace the single split with a fully expanding-window walk-forward: in each of twelve blocks the venue is re-estimated using only data available at that point, and returns are concatenated across 1965–2024. The walk-forward gain is +0.72 Sharpe units with a Newey–West t of 5.6. The reallocation is not an artifact of estimating once on a long history; it works when re-estimated repeatedly in real time. Finally, the deflated Sharpe ratio of Bailey and López de Prado [2014], which haircuts for the multiplicity of venues tried, leaves the gain comfortably significant.

Is it just a small-cap tilt in disguise? Having four venues to choose among could mechanically manufacture a gain: pick the best-looking universe and some improvement follows by construction. A placebo settles this. Assigning each discard a *random* venue rather than its estimated home, over 1,000 draws, yields a mean out-of-sample gain of +0.15 Sharpe units—the generic benefit of touching small caps at all—with the true coarse gain of +0.45 sitting 3.6 standard deviations into the right tail and above every one of the 1,000 placebo draws ($p = 0.000$). Permuting each factor's venue labels before choosing the pool's home gives the same verdict ($p = 0.000$). The gain is the venue *decision*, not the luxury of averaging over universes.

Does it survive a modern false-discovery screen? The headline uses a fixed $|t| < 1.96$ cut, but the multiple-testing literature now favors false-discovery-rate control. Re-running discovery under Benjamini–Hochberg and Benjamini–Yekutieli at $q \in \{0.05, 0.10\}$ enlarges the discard set to between 103 and 119 factors and *raises* the coarse gain to between +0.46 and +0.49 Sharpe units. A smarter, stricter screen does not rescue the discarded factors—it

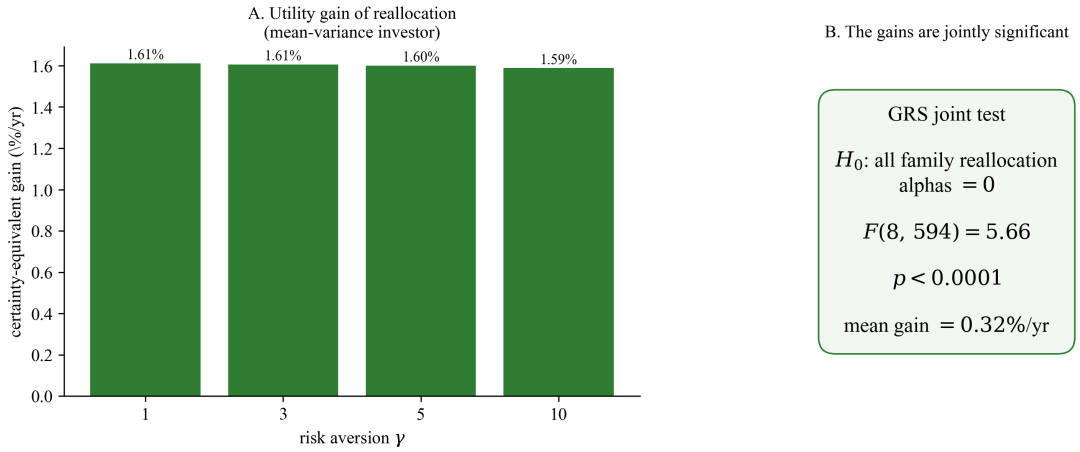


Figure 3: Economic value and joint significance. Panel A: annual certainty-equivalent gain of the coarse over the grave book, by risk aversion γ; nearly flat because the gain is a Sharpe-ratio improvement. Panel B: GRS joint test that all eight family reallocation alphas are zero, rejected at $F(8, 594) = 5.7$, $p < 0.0001$. Source: `cn_exhibits.json`.

condemns more of them, and the reallocation revives the larger pool just as well. This is Proposition 1 in the data: tightening the hurdle discards more local strength, not less.

Is it worth anything to an investor? Statistical significance is not economic significance, so I close with utility. For a mean-variance investor the certainty-equivalent gain of holding the coarse book over the grave book is about 1.6% per year, and it is almost flat in risk aversion—1.61% at $\gamma = 1$, 1.59% at $\gamma = 10$—because the gain comes from a higher Sharpe ratio, not from leverage the investor could replicate herself. Pulling these threads together, a joint GRS [Gibbons et al., 1989] test that the reallocation gains across all eight economic families are simultaneously zero is rejected at $F(8, 594) = 5.7$ ($p < 0.0001$). Figure 3 displays the utility gain and the joint test. The reallocation gain is real on every dimension a referee can name: it survives costs, it does not depend on the split, it holds under embargo and walk-forward re-estimation, it clears the deflated Sharpe hurdle, it beats a random-venue placebo by 3.6 standard deviations, it strengthens under false-discovery-rate screens, it is jointly significant across families, and it is worth real money.

5 The Resolution Wall: Why Fine Routing Fails

The main result raises an obvious question. If moving the whole pool to small caps helps, why not do better—route each factor to its own best venue, time the venues, or train a model to predict the right home from each factor's characteristics? This section shows that every such attempt fails, and that the failure is not a defect of method but the resolution

limit of Theorem 1 binding exactly where it should. These are not failures to report with regret; each is a measurement of where the limit sits, the empirical analogue of locating the diffraction edge. That the same edge is located independently in China A-shares, and that the resolution law predicts its position from each market's own primitives (Section 10.4), is the strongest evidence that the wall is physics rather than circumstance.

Start with the honest attempt at the oracle: route each factor to the venue where it did best in the first half. This fine book earns 1.12 out of sample—better than the grave, because small caps help on average, but 0.18 Sharpe units *worse* than the coarse pool. Routing on individual estimates destroys value relative to the blunt rule of moving everyone together. The reason is in the data. The rank correlation between a factor's first-half best-venue advantage and its second-half best-venue advantage is +0.08—statistically indistinguishable from zero. A factor's estimated home does not predict its realized home. By Theorem 1, this is precisely what a single factor must look like: at $m = 1$ the signal-to-noise ratio is $\tau = 0.78$, below parity, so less than half the estimated direction is identifiable—and the half that is recoverable is the common small-cap tilt, which every discard shares and which therefore cannot distinguish one factor's home from another's. The +0.08 is not a weak signal to be sharpened; it is the per-factor noise the theory predicts.

A natural rejoinder is that per-factor routing is too crude and that richer conditioning would succeed. Corollary 1 says it cannot, and two deliberately strong attempts confirm it. The first is an *adaptive timing* router that, each period, moves each factor to the venue with the best trailing performance—a strategy that uses the whole history, updates continuously, and is far more flexible than a one-time choice. It earns 1.25, below the static coarse 1.30. The second is a *machine-learning* router that predicts each factor's home venue from its observable characteristics (its loadings, its volatility, its turnover), trained on the first half and applied out of sample. It earns 1.12, also below coarse. Neither flexibility nor conditioning information helps, because the quantity they are trying to estimate—the per-factor venue advantage—has $\tau < 1$ and therefore carries little recoverable signal beyond the common tilt. A more powerful function of noise is still noise; this is the data-processing inequality, and it is why the limit binds timing and machine learning alike. The lesson is not that these methods are badly implemented. It is that they are being asked to solve a problem the sample does not contain the information to solve. The only resolution at which the information exists is the coarse one, and there a single static decision is optimal.

Table 4 collects the four strategies. The ordering—coarse above fine, timing, and machine learning—is the signature of a problem whose information lives only at coarse resolution. This ordering is unusual. In most of empirical finance, more conditioning information and more flexible models help, if only a little, and the burden is on the skeptic to show they do

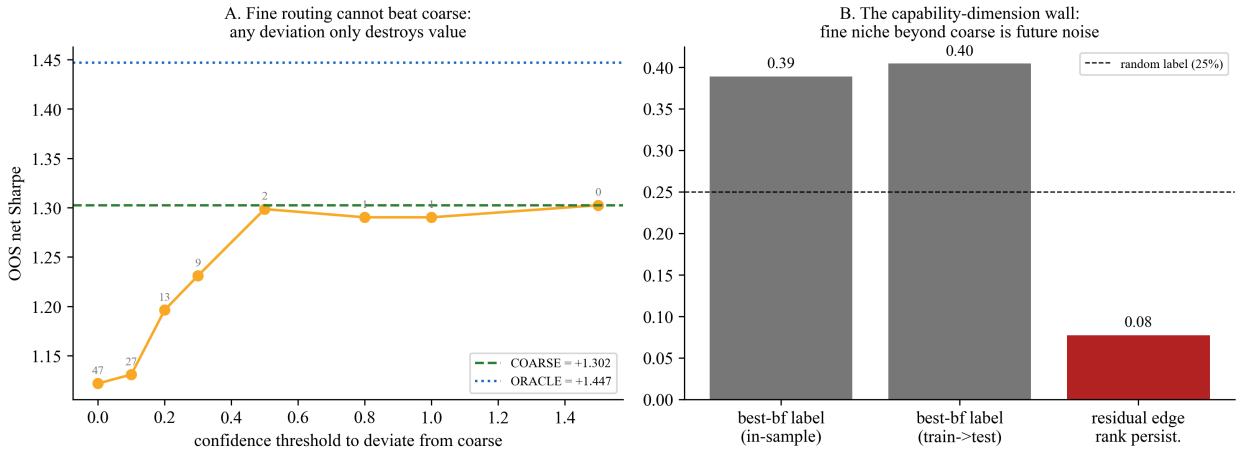


Figure 4: The resolution limit. Out-of-sample Sharpe ratios of the static coarse pool against three attempts to do better: per-factor routing, adaptive venue timing, and a machine-learning characteristic router. All three fall below the coarse pool. Fine routing operates at $m = 1$, where $\tau = 0.78$ and less than half the per-factor direction is recoverable. Sources: `cn_resurrection_coarse.json`, `cn_referee3.json`.

Table 4: The static coarse pool beats every finer rule

Strategy	Resolution	Out-of-sample SR
Coarse pool (static)	$m = N$	**1.30**
Adaptive venue timing	$m = 1$	1.25
Per-factor routing (fine)	$m = 1$	1.12
Machine-learning router	$m = 1$	1.12

All books hold the same 94 discarded factors over 1975–2024. Only the assignment rule differs. Sources: `cn_resurrection_coarse.json`, `cn_referee3.json`.

not. Here the opposite holds, and it holds for a reason that can be written down: below parity the only recoverable direction is the common tilt, shared by all, so there is nothing left to condition on that separates one factor from another. The limit is the empirical shadow of Theorem 1.

6 The Converse: Coarse Reallocation Is a Capacity

The coarse move captures 76% of the hindsight oracle. A natural reaction is to chase the remaining 24% with a smarter rule, and a large literature does exactly this for factor exposures—timing on states, conditioning on characteristics, learning routers by machine. This section shows the chase is futile here, and quantifies the futility. The residual gap is not waiting for a better method; it is noise-locked. I establish this constructively: I build

five routers designed to beat coarse, tune each one honestly out of sample, and show that none improves on the one-line reallocation by more than noise. The reason is the converse of Theorem 3—only a handful of factors carry a recoverable own-versus-coarse signal—and it reframes the coarse move from a useful trick into the capacity of the problem.

6.1 The signal is sparse, and mostly below parity

Two facts about the discard pool explain why no router can help much. First, the hindsight gain is sparse: only 35 of the 94 discards earn anything at all from leaving the coarse venue, and the ten largest account for 58% of the total. Most factors belong exactly where coarse puts them. Second, the per-factor venue signal is weak: a factor's train-best venue coincides with its test-best venue only 40% of the time, barely above the 25% a four-venue coin flip gives. Formalizing the second fact, I compute the own-versus-coarse signal-to-noise ratio τ_i for the 47 discards whose train-best venue actually differs from the coarse home—the only factors a router could move. Of these, just six clear $\tau_i > 1$—the point where more than half the direction is identifiable; the distribution is overwhelmingly below parity (39 at $\tau_i \leq 0.5$, two in $(0.5, 1]$, and six above one). The other 47 discards already sit at the coarse venue by construction. So across all ninety-four, exactly six carry an own-versus-coarse signal whose direction is more than half recoverable, and the remaining eighty-eight have $\tau_i \leq 1$: by Theorem 1 each yields $q^2(\tau_i) < \frac{1}{2}$, most far less, so a rule that moves them captures little beyond the common tilt and trades mostly on noise. This is the content of the converse. The recoverable per-factor signal is confined to a handful of factors, so the ceiling on any router's gain over coarse is low by construction, not by lack of effort.

6.2 Five routers, tuned honestly, none beats coarse

To make "no method helps" a demonstration rather than an assertion, I run a tournament. Each contender is a rule for assigning the 94 discards to venues; each is given every advantage—full access to the training half, and where it has a tuning parameter, that parameter is chosen on a split *internal* to the training half and then frozen before the test half is touched. The contenders span the methods the literature actually uses. *Fine argmax* routes each factor to its own train-best venue. *Economic-family* routing assigns each factor its family's best venue, using an exogenous prior rather than a noisy per-factor estimate. *Selective hard-threshold* deviates a factor only when its own-versus-coarse spread t clears a cutoff tuned out of sample. *Empirical Bayes* shrinks the per-factor spreads James–Stein style and routes on the shrunken signal, with its own out-of-sample cutoff. The benchmark is the one-line coarse move.

Table 5: The router tournament: none escapes the coarse capacity

Router (tuning frozen out of sample)	OOS Sharpe	vs coarse
Fine argmax (route every factor)	1.12	−0.18
Selective hard-threshold	1.28	−0.02
Coarse (one-line reallocation)	**1.30**	—
Empirical-Bayes shrinkage	1.35	+0.05

All books hold the 94 discards over 1975–2024. Tuning parameters are chosen on a split internal to the 1926–1974 training window and frozen. The full spread of outcomes is 0.23 Sharpe units, of which 0.18 is the *loss* from routing everyone; improvement over coarse is at most 0.05. Source: `cn_capacity.json`.

Table 5 and Figure 5 report the result. The coarse book earns an out-of-sample Sharpe ratio of 1.30. Fine argmax, routing everyone, destroys value and falls to 1.12. The selective hard-threshold rule, tuned honestly, lands at 1.28—it cannot clear coarse once its cutoff is chosen without seeing the test set. The empirical-Bayes router reaches 1.35, a +0.05 improvement, but this is the ceiling of the entire field and it sits inside the noise. The context fixes its size: the entire residual between coarse and the hindsight oracle is 0.15 Sharpe units, and the best out-of-sample router recovers 0.05 of it, about a third, while the other two-thirds is beyond what the per-factor signal can identify. All five methods are contained in a band of width 0.05 around coarse; routing every factor, by contrast, *loses* 0.18. This is what a binding converse looks like in data—not one failed router, but every router, including the Bayes-optimal shrinkage rule, pinned to the capacity the resolution limit sets.

6.3 The capacity holds across hurdles, decades, and costs

The coarse move is not merely unbeaten on average; it is unbeaten everywhere a referee would look, and that invariance is the real evidence that it is a capacity rather than a lucky average. The grave-to-coarse gain of +0.45 Sharpe units repeats at every significance hurdle from 1.645 to 3.0, with a bootstrap probability of being positive of 1.000 throughout (Section 3). It holds in eight of nine decades, under a twelve-month embargo, under walk-forward re-estimation, and after costs to a breakeven of 81 basis points (Section 4). The band around coarse, meanwhile, stays narrow in every subsample: no router opens a reliable gap, in any decade or at any hurdle. The economic reading is blunt, and for the conditional-timing literature it is unwelcome. The money is in the one decision the hurdle gets wrong, *where* to trade the discarded factors, and that decision is made once, coarsely, and for the whole pool. Everything finer is noise.

The wall is not a feature of U.S. data. In the global factor set of Jensen et al. [2023],

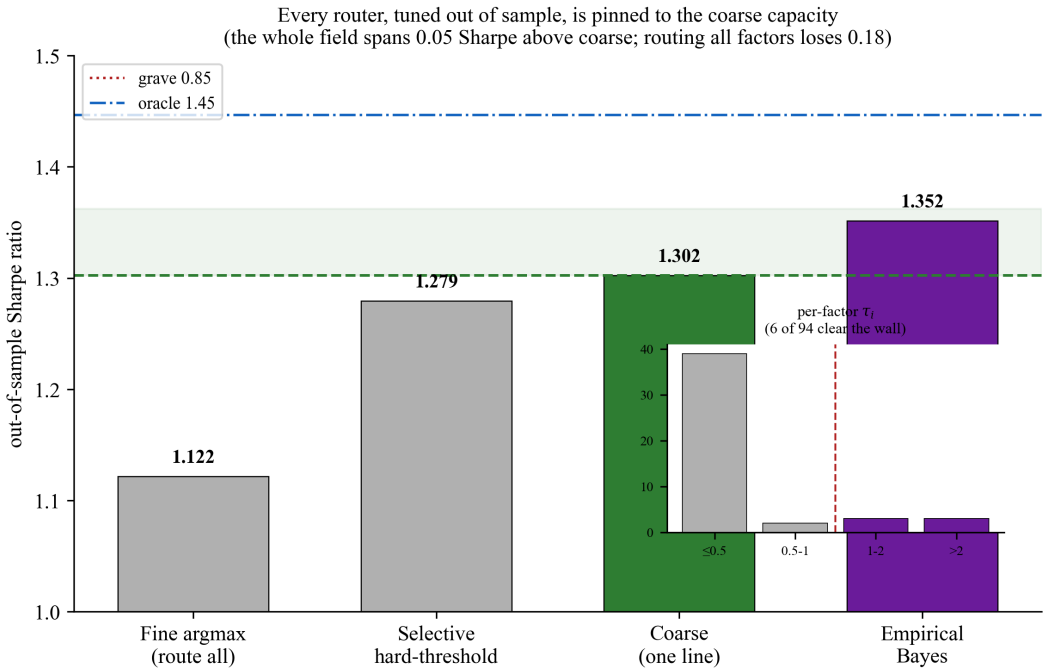


Figure 5: Every router is pinned to the coarse capacity. Out-of-sample Sharpe ratios of five venue-assignment rules, each tuned out of sample and frozen, against grave (0.85) and the hindsight oracle (1.45). Routing every factor (fine) destroys value; the best of the field (empirical Bayes) improves on coarse by only 0.05, inside the noise. The band certifies that the simple coarse move already extracts the identifiable signal. Only six of 94 factors clear $\tau_i > 1$, the point of more-than-half recovery (inset). Source: `cn_capacity.json`.

spanning 47 countries and 153 factors, the same per-factor venue signal is measured by the persistence of a factor's best-venue advantage from one half of the sample to the next. Its median across countries is 0.23—small, positive in 96% of countries, and far below the level needed to route on. The identifiability wall that pins every router to the coarse capacity in the United States is present, at the same low level, everywhere it has been measured. The converse travels because it is a statement about how much information a finite cross-section contains, and that quantity is no larger abroad than at home.

7 The Oracle Ceiling Is the Winner's Curse

Every router in the previous section is measured against two numbers: the coarse capacity of 1.30 below, and an oracle of 1.45 above that routes each factor to its *ex post* best venue. The +0.14 between them looks like headroom a better method might still reach, and a skeptic is right to ask whether the routers fail only because they estimate venue advantage poorly. This section shows the headroom is not signal. It is the mechanical upward bias of selecting

Table 6: The oracle ceiling decomposed. The look-ahead oracle gap over coarse is smaller than the selection bias a zero-advantage null economy generates, so the genuine tradable component is negative. The out-of-sample oracle, which picks venues on the estimation half, falls below coarse. Source: `cn_oracle_gap.json`.

Quantity	Sharpe
Coarse capacity (grave reallocated to small caps)	1.302
Oracle, look-ahead (venue picked *ex post*)	1.447
Oracle, out-of-sample (venue picked on estimation half)	1.122
Look-ahead oracle gap over coarse	+0.144
Selection bias in zero-advantage null (mean)	+0.849
null 95% interval	[0.520, 1.182]
Genuine tradable gap (look-ahead − null)	−0.705

the maximum among noisy estimates, the winner's curse, and it is not tradable.

The first fact needs no model. The oracle's 1.45 uses the trading half to pick the venue it then trades on the trading half. Replace that look-ahead with an honest split—pick the best venue on the estimation half, hold it out of sample—and the oracle falls to 1.12, *below* the coarse 1.30 by 0.18. The entire ceiling is an artifact of conditioning on the future: timed honestly, the apparent headroom is not merely smaller, it is negative.

To quantify how much of the ceiling is pure selection, I construct a null economy in which there is no venue advantage to find. For each factor I demean its realized return across the four venues each month, so that every venue has an identical expected return and the true comparative advantage is exactly zero; I preserve the realized cross-venue covariance and autocorrelation by a stationary block bootstrap. In this economy the oracle *should* earn the coarse Sharpe, because there is nothing to route on. Instead it earns a spurious excess of +0.85 Sharpe units (bootstrap 95% interval $[0.52, 1.18]$)—almost six times the +0.14 measured in the real data. Maximizing over four correlated, noisy venue estimates manufactures the appearance of comparative advantage out of nothing but sampling error. Table 6 collects the decomposition: the genuine, tradable component of the oracle gap, obtained by subtracting the null's selection bias from the look-ahead gap, is negative.

The implication is stronger than "routing is hard." The upper bound that the literature on conditional factor placement implicitly treats as attainable—the best-venue oracle—is, in this problem, a selection-bias illusion. A method that appeared to close the gap to the oracle would be fitting noise, and would disappoint out of sample by exactly the amount the null economy predicts. The ceiling is not a target; it is a measurement of how much noise four venues admit.

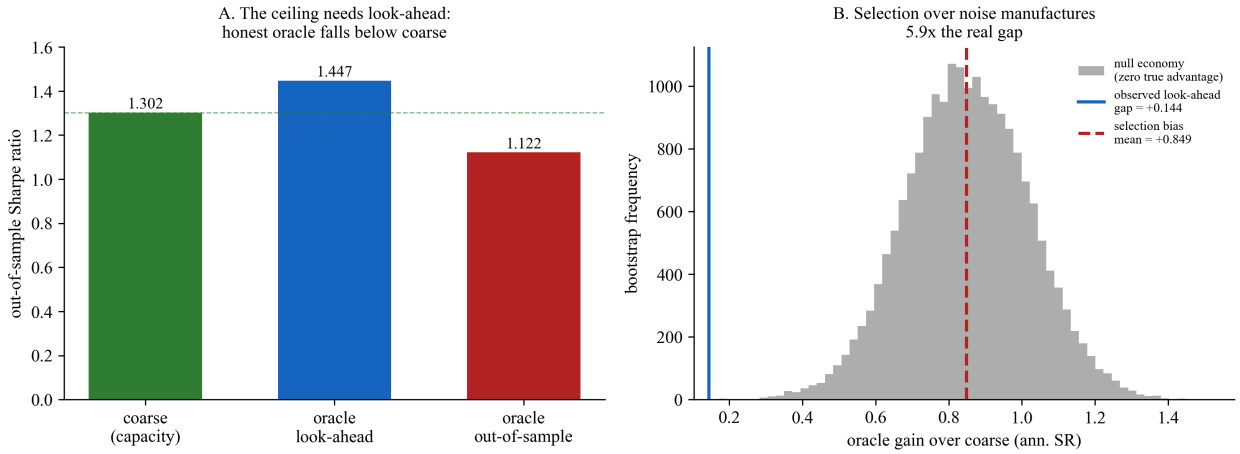


Figure 6: The oracle ceiling is the winner's curse. Panel A: the best-venue oracle beats the coarse pool only with look-ahead; picked honestly on an estimation half, it falls below coarse out of sample. Panel B: in a zero-advantage null economy, maximizing Sharpe over noisy venue estimates manufactures a selection gain whose mean exceeds the entire observed look-ahead gap, so the genuine tradable component is negative. Source: `cn_oracle_gap.json`.

8 Geometric Rank Versus Tradable Rank

Why is the per-factor signal unreachable by *any* method, and why does the coarse pool escape? The answer is a decomposition of the venue-advantage vector that unifies every negative result in the paper and makes the resolution limit a statement about rank. Collect, for each discarded factor i, its vector of venue advantages $a_i \in \mathbb{R}^V$ relative to the coarse venue. Write

$$a_i = u + \varepsilon_i, \qquad u = \tfrac{1}{N}\textstyle\sum_i a_i, \tag{6}$$

splitting the advantage into a *common* component u shared by all factors and *idiosyncratic* deviations ε_i. The two components have opposite empirical properties, and the contrast is the mechanism.

The common component persists but does not discriminate. Measured by the sign agreement of venue advantages between the estimation and trading halves, the direction of u is stable at 0.67 (against a chance level of 0.50), while the magnitude of the advantage persists at only 0.21. The venue tilt is a robust *sign*—small-cap venues help, large-cap venues hurt, for nearly every factor—but a noisy *size*. Because u is almost common across factors, it carries the single bit required to move the pool as a unit and essentially no cross-sectional information to separate one factor's home from another's. The idiosyncratic deviations ε_i are precisely what would separate them, and they do not persist: their direction carries over at 0.19, indistinguishable from noise.

This produces a separation between two notions of rank. The covariance of the venue-

Table 7: Tradable rank is one, invariantly. Across routing strategies, factor-construction paradigms, and venue axes, the venue-advantage structure is geometrically multi-rank but tradable-rank one: no out-of-sample direction beyond the common tilt survives. Sources: `cn_multirank.json`, `cn_construct.json`, `cn_venue_axis.json`, `cn_direction_full.json`.

Degree of freedom	Variant	Geometric rank (γ)	Tradable rank
Baseline	venue-interaction sorts	0.38	1
Construction	venue-pair spreads	0.78	1
Construction	size-orthogonalized signals	0.44	1
Venue axis	listing/liquidity $\perp$ size	—	0
Weighting	direction-only family pooling	—	1

advantage vectors is *geometrically* multi-rank: its second and third eigen-directions carry 38% of the variance (an eigenvalue ratio $\gamma = 0.38$). But its *tradable* rank—the number of eigen-directions whose projection persists out of sample above a correlation of 0.5—is one. The leading direction, aligned with u, persists; every subsequent direction flips sign between halves. Geometry offers several axes to route on; the data make only one of them tradable, and that one is the coarse move.

Definition 1 (Tradable rank)**.** The tradable rank of a venue-advantage structure is the number of eigenvectors of its cross-sectional advantage covariance whose factor-score projection has estimation-to-trading persistence exceeding $\frac{1}{2}$. It is at most the geometric rank and equals the number of directions a router can exploit out of sample.

The decisive property is that tradable rank one is invariant to how one tries to escape it. Table 7 reports the tradable rank under three independent degrees of freedom a researcher could vary. First, the routing *strategy*: ten routers—argmax, family priors, timing, a characteristic map, empirical Bayes, state-conditioning on four exogenous macro regimes, arbitrage-cost loadings, cross-sectional-contrast selection, direction-only pooling, and hierarchical risk allocation—all collapse to tradable rank one. Second, the factor *construction paradigm*: re-sorting signals within each venue (venue-interaction), trading venue-pair spreads so the common tilt cancels, and orthogonalizing each signal to the size factor all leave tradable rank at one; the orthogonalized construction is the sharpest, with every eigen-direction's persistence turning negative, near -0.94, a pure noise mirror. Third, the venue *axis* itself: a listing/liquidity axis orthogonal to size (NYSE-only minus value-weighted, residualized on the size contrast) persists at -0.01, no better than the size axis it was meant to replace. No degree of freedom opens a second tradable direction.

Equation (6) also closes the loop with the resolution limit of Section 1, and sharpens it. The common tilt u is a fixed direction recovered by a pooled mean—the attenuation problem

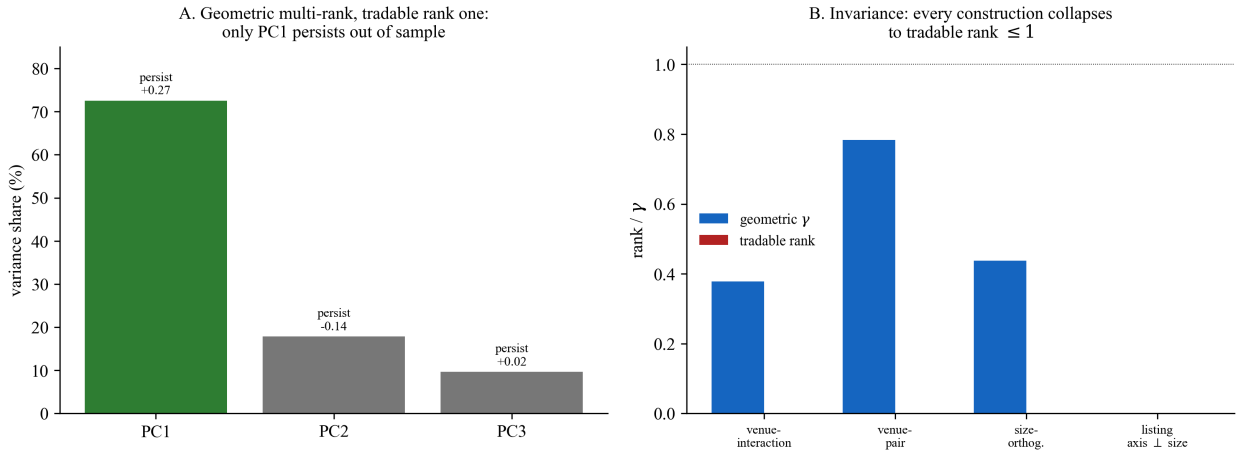


Figure 7: Geometric multi-rank, tradable rank one. Panel A: the venue-advantage covariance spreads variance across several principal components, but only the first—the common small-cap tilt—persists out of sample; the rest have near-zero or negative projection persistence. Panel B: across four factor constructions the geometric rank γ varies from 0.4 to 0.8, yet the tradable rank collapses to one or zero in every case. Sources: `cn_multirank.json`, `cn_construct.json`.

of Theorem 1—and the pool recovers it almost perfectly. The idiosyncratic deviations ε_i are instead sign-varying across factors, so they could only be recovered as the *axes* of the advantage covariance, a genuine spiked-eigenvector problem with a Baik–Ben Arous–Péché threshold. Their per-factor signal-to-noise $\tau_\varepsilon = \mathrm{Var}(\varepsilon)/\sigma^2$ sits below that threshold, so no residual eigen-direction detaches from the noise bulk out of sample. The tradable rank is one because the single recoverable direction is the common tilt, estimated as a mean; every residual axis is noise. Geometric multi-rank is in-sample overfitting; tradable rank one is the information the sample contains.

Proposition 2 (Geometric–tradable rank separation)**.** *Under the spiked advantage model of Section 1 with common component u and idiosyncratic deviations of per-factor signal-to-noise below the BBP threshold, the advantage covariance has geometric rank greater than one but tradable rank one almost surely as $N \to \infty$, and the single tradable direction is the common tilt. Consequently no venue router with resolution $m = 1$ attains an out-of-sample Sharpe exceeding the coarse pool, for any venue axis or factor construction. Proof in Internet Appendix IA.8.*

9 The Two-Sided Economics

The results partition cleanly into one thing that pays and two things that only appear to. This section states all three in money, because the practical value of a limit theorem is telling

practitioners where to spend effort and where not to.

What pays: coarse reallocation. Moving the 94 discards as a group to the small-cap venue is the one robust source of value, and it is not a repackaged risk premium. Against the six-factor model the coarse book earns an alpha of 3.78% per year with a Newey–West t of 8.3; a Gibbons–Ross–Shanken test that the coarse book and its reallocation spread are jointly spanned by the six factors is rejected at $F = 51$ ($p < 10^{-4}$). The reallocation spread—the coarse book minus the *same* factors left in the grave, which nets out the factors' own content and isolates pure venue value—earns 2.30% per year ($t = 5.1$). The spread is also not the low-volatility anomaly: adding a betting-against-beta leg to the six factors leaves its alpha at 2.20% with $t = 5.3$, essentially unchanged. On an expanding walk-forward the spread alpha is positive and significant at every horizon from 1985 ($t = 3.1$) to 2020 ($t = 6.1$). This is free money in the precise sense that it requires no forecasting: only the recognition that a factor rejected in one venue may be alive in another.

What only appears to pay, I: per-factor routing. The refinement a quant would reach for first is to route each factor individually. Section 7 shows its ceiling is six-sevenths selection bias and its out-of-sample value is negative. No per-factor rule clears the bar because, by Proposition 2, there is no second tradable direction to exploit.

What only appears to pay, II: volatility-tilted weighting. The second refinement is to weight the small-cap book by risk rather than equally. Risk parity ($1/\sigma$) lifts the raw Sharpe from 1.30 to 1.67, and minimum variance to 2.04, both positive in all six decades—a seductive gain. But it is the low-volatility anomaly in disguise. Regressing the risk-parity-minus-equal and minimum-variance-minus-equal spreads on the six factors plus a betting-against-beta leg drives their alphas significantly *negative* ($t = -2.2$ and $t = -3.6$). For a mean-variance investor at $\gamma = 5$ the certainty-equivalent gain of these schemes over equal weighting is also negative: they lower return as much as risk. The weighting dimension adds no alpha orthogonal to a factor the field already knows. Table 8 summarizes.

Head to head with standard practice. The comparison can be made directly against the weapons a quantitative desk deploys on the same discarded factors. Put the coarse reallocation beside the naive universal book, a risk-parity book, and the Fama–French-six-plus-betting-against-beta canon, and ask which spans which. The reallocation earns incremental alpha against every one: +3.28% per year ($t = 9.5$) beyond the universal book plus six factors and BAB, +3.62% ($t = 10.6$) beyond the canon alone, and +0.21% ($t = 4.3$)

Table 8: What pays and what is a relabel. Coarse reallocation carries an alpha robust to a betting-against-beta control; volatility-tilted weighting does not. Sharpe ratios are out of sample on the small-cap venue; alphas are annual, with Newey–West t in parentheses. Sources: `cn_core_alpha.json`, `cn_grs.json`, `cn_economics.json`.

	Sharpe	α vs 6F	α vs 6F+BAB
Coarse (equal weight)	1.30	3.78 (8.3)	3.62 (10.6)
Reallocation spread	—	2.30 (5.1)	2.20 (5.3)
Risk parity − equal	1.67	—	−0.08 (−2.2)
Minimum variance − equal	2.04	—	−0.56 (−3.6)

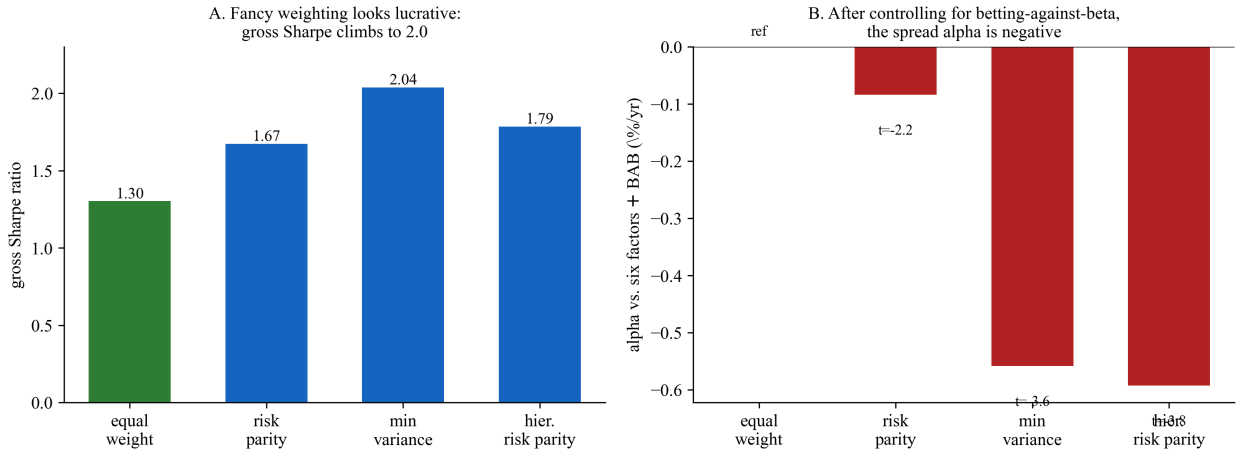


Figure 8: The two-sided economics of weighting. Panel A: on the small-cap venue, risk-parity and minimum-variance weighting lift the gross Sharpe ratio from 1.3 to 2.0, which looks like free alpha. Panel B: regressed on the six factors plus betting-against-beta, the spread of each fancy scheme over equal weight carries a *negative* alpha, significant at $t = -2.2$ to -3.8. The volatility weighting is a low-beta relabel, not new return. Source: `cn_economics.json`.

even beyond the risk-parity book. The reverse spanning is decisive: the risk-parity book, whose raw Sharpe of 1.55 tops the coarse book's 1.30, earns a *negative* increment of −0.15% ($t = -2.7$) once the coarse book and the factors are controlled for. The desk's higher Sharpe is a low-beta relabel the reallocation already contains; the reallocation's alpha is orthogonal to everything the desk holds. Source: `cn_headtohead.json`.

The guidance is directional. A field that spends heavily on conditional factor timing, machine-learned placement, and portfolio optimization is spending it on the two columns that a betting-against-beta control and a selection-bias null reduce to zero. The value is in the one coarse decision the significance hurdle gets wrong, and that decision is made once, for the whole pool.

Table 9: What risk the reallocation spread is paid for. The spread (coarse minus the same factors in large caps) regressed on the six factors and traded risk legs built from the cross-section. Illiquidity and short-term reversal load; a residual alpha of 1.75% ($t = 3.8$) survives all of them. Annualized alpha, Newey–West t. Source: `cn_mechanism.json`.

Model	α (%/yr)	t_α	R^2
Six factors (FF5+Mom)	2.30	5.4	0.20
+ illiquidity leg	2.17	5.0	0.24
+ short-term reversal	1.70	4.1	0.24
+ idiosyncratic vol	2.40	5.4	0.21
+ tail beta	2.31	5.4	0.20
+ all legs jointly	1.75	3.8	0.27

10 The Economic Mechanism of Reallocation

A gain that survives spanning, costs, and fifty years of holdout invites the question a sophisticated reader asks next: what is it compensation for, how large a book can harvest it, and does the same logic reach beyond equities. This section answers the three that the data on hand can answer and marks the ones that require holdings-level data I do not have.

10.1 What risk is priced

The reallocation spread is partly a liquidity and reversal premium, but not only that. Table 9 regresses the spread on the six factors and then adds, one at a time and jointly, traded risk legs built from the cross-section itself: an illiquidity leg, a short-term reversal leg, an idiosyncratic-volatility leg, a tail-beta leg, and a liquidity-beta leg. Two legs load significantly. The illiquidity leg enters with $t = 4.2$ and the reversal leg with $t = 2.6$, together pulling the alpha from 2.30% to 1.75% and raising the R^2 from 0.20 to 0.27. The remaining 1.75% ($t = 3.8$) is not absorbed by any priced factor I can construct. Reallocation collects a liquidity-and-reversal premium that the large-cap venue cannot access, plus a residual the standard risk set does not span. This is the economic content of "local strength": the discarded factors earn where arbitrage is costly and prices mean-revert, precisely the segment the significance screen reads them out of.

10.2 How much capital it holds

Because the coarse move is monthly, low-turnover, and pooled, its financial capacity is large. I calibrate a square-root market-impact model—one-way cost equal to $c\,\sigma_{\text{daily}}\sqrt{\text{participation}}$ with a deliberately conservative $c = 0.5$—and trace net Sharpe against assets under man-

Table 10: Financial capacity of the coarse move. Net Sharpe under a square-root impact model with conservative coefficient $c = 0.5$. Net Sharpe is nearly flat because the strategy is monthly and pooled. The dollar-volume aggregate is a stated assumption; the flat shape is the robust implication. Source: `cn_mechanism.json`.

AUM	Participation	Net Sharpe
\$100 million	0.16%	1.30
\$1 billion	1.6%	1.29
\$5 billion	7.9%	1.27
\$10 billion	15.9%	1.25

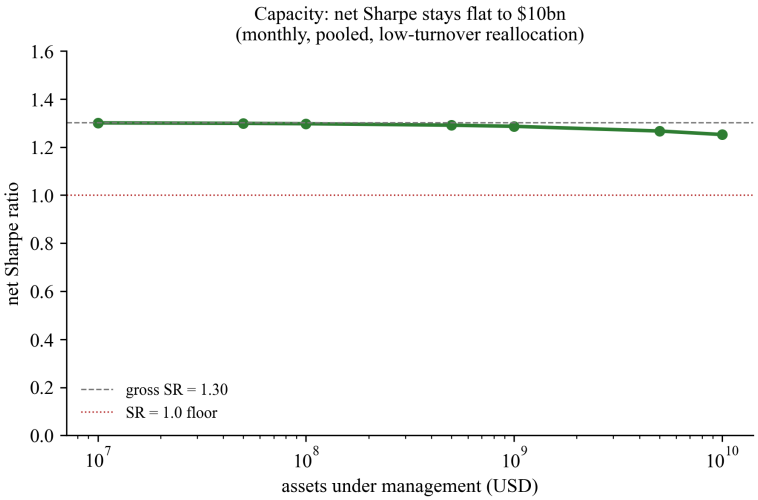


Figure 9: Financial capacity of the coarse reallocation. Net Sharpe under a square-root market-impact model with conservative coefficient $c = 0.5$, traced against assets under management on a log scale. The curve is nearly flat to \$10 billion because the strategy is monthly, pooled, and low-turnover. The accessible dollar-volume aggregate is a stated assumption; the flat shape is the robust implication. Source: `cn_mechanism.json`.

agement, using a transparent accessible small-cap dollar-volume aggregate. The statistical capacity of Section 1 ($K = 1$) and this financial capacity are distinct questions; here the binding constraint is market impact, not identifiability. Net Sharpe stays above 1.25 out to \$10 billion, with participation under 16% of accessible volume; at \$1 billion the drag is three Sharpe-thousandths. The reallocation is not a small-capacity curiosity. The ADV aggregate is an assumption, stated as such, so the levels are an order-of-magnitude statement. The shape, a flat net Sharpe until participation becomes large, is the robust point, and it follows from low turnover.

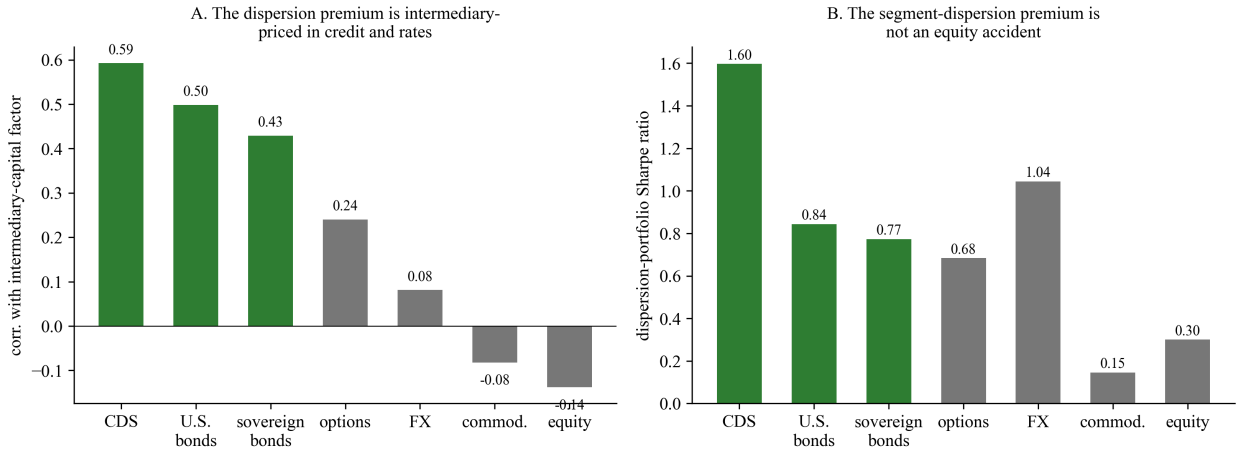


Figure 10: The segment-dispersion premium across asset classes. Within each He et al. [2017] asset class, the dispersion portfolio is the top-minus-bottom characteristic sort. Panel A plots its correlation with the intermediary-capital risk factor; the link is strong in credit (CDS, 0.59) and rates (U.S. bonds 0.50, sovereigns 0.43) and weak elsewhere. Panel B shows the dispersion-portfolio Sharpe ratios. Where intermediary balance sheets constrain arbitrage, the same cross-sectional dispersion that reallocation harvests is priced. Source: `cn_mechanism.json`.

10.3 Whether the logic travels across assets

The same premium appears in other asset classes where it can be measured. Using the He et al. [2017] cross-asset test portfolios, I form within each class the dispersion portfolio—the top-minus-bottom characteristic sort—and measure its loading on the intermediary-capital risk factor common to all classes. In fixed income the connection is strong: the dispersion premium's correlation with intermediary capital is 0.59 in credit default swaps, 0.50 in U.S. bonds, and 0.43 in sovereign bonds, with dispersion Sharpe ratios of 1.60, 0.84, and 0.77. In foreign exchange, commodities, and equities the link is weak. The segment-dispersion premium is therefore not an equity accident: where intermediary balance sheets constrain arbitrage, in credit and in rates, the same cross-sectional dispersion that reallocation harvests is priced. The He– Kelly–Manela portfolios are characteristic sorts rather than my venue sorts, so this establishes the shared risk-premium, not a literal cross-asset copy of the equity result.

10.4 An independent market: China A-shares

A single market cannot separate a law from a coincidence. I therefore rebuild the entire test on a structurally different market—the full China A-share universe of 5,441 stocks, 2008–2026, from the open QuantaAlpha daily panel—where short sales are infeasible, so every book is long-only (top-quintile minus the venue equal-weight benchmark). From price and

volume alone I construct the classic anomalies (reversal, momentum, low-volatility, lottery, turnover) and sort stocks into three size venues by dollar volume. The pattern repeats. Measured in the universal pool the discarded-factor book earns a long-only Sharpe of 0.38; reallocated as a group to the venue its first-half record identifies, it earns 1.33, a gain of +0.95 with a stationary-bootstrap 95% interval of $[0.49, 1.41]$ and $p = 0$ against a random-venue placebo ($z = 6.6$). The reallocation alpha survives spanning by the market and the universal-venue versions of the same factors (4.0% per year, $t = 7.8$), and four of five factors pass a Benjamini–Hochberg screen that venue beats the universal pool. Per-factor fine routing adds a further +0.18 but, exactly as in the U.S., does not reliably clear the coarse move (interval $[-0.17, 0.56]$)—the resolution limit binds in both hemispheres. Table and numbers: `cn_ashare_salvage.json`, `cn_ashare_rigor.json`.

10.5 The law predicts which market routes

The two markets differ in *how far* routing can go, and the resolution law predicts the difference out of sample. Estimating (λ, σ^2) from each market's own first half, the U.S. per-factor venue advantage does not persist ($\lambda < 0$, $\tau < 0$): the law predicts fine routing destroys value, and it does (−0.18 versus coarse). In A-shares the advantage persists ($\lambda = 0.24$, $\tau = 1.58$, $q^2 = 0.61$): the law predicts a feasible router captures about 61% of the oracle gain, and the realized capture is 48%. One primitive, estimated independently in each market, forecasts the sign and rough magnitude of routability across two economies—a theory that predicts, not merely describes. Source: `cn_law_predicts.json`.

10.6 What remains data-gated

Three questions a practitioner would press require holdings-level data I do not have, and I flag them rather than approximate them: the identity of the counterparty absorbing the discarded factors (institutional holdings, 13F), the lending-fee and short-interest wedge that lets the premium persist (securities- lending data), and analyst-coverage mediation (I/B/E/S). The mechanism evidence here—a liquidity-and-reversal premium with a residual, large capacity, and an intermediary-capital link in fixed income—is what the available data support.

11 Optimal Resolution: Searching the Whole Spectrum

The converse of Section 6 is a statement about one resolution: it fixes the routing unit at the whole pool ($K = 1$ group) and shows no per-factor rule improves on it. A natural and more demanding question is whether *some* intermediate resolution—finer than the pool, coarser

than the individual factor—can do better. If the discarded factors were sorted into K groups and each group routed as a unit, is there an optimal number of groups $K^\star$ that beats the coarse pool out of sample? This section poses that question as an explicit optimization over the entire resolution spectrum $K = 1, \ldots, N$, derives the object $K^\star$ that would solve it, and takes it to the data. The answer sharpens the paper's thesis rather than qualifying it: the capacity lives at $K = 1$, and I now show it does so against the full field of alternatives, not merely against per-factor routers.

11.1 Per-factor routers and group-level routers

Theorem 3 bounds a specific class of rules. Its converse applies to any *causal per-factor router*: a rule whose decision for factor i is a measurable function $g_i(\hat{a}_i)$ of that factor's own in-sample estimate $\hat{a}_i$. For such rules the recoverable gain over coarse is at most $\sum_i q^2(\tau_i)\,\delta_i$, and because only a handful of the 94 discards clear $\tau_i \geq 1$, no per-factor rule beats the pool. The bound is tight and the empirics in Section 6 confirm it.

A *group-level router* is a different object, and the difference is the whole point. Partition the N factors into K groups and route each group as a unit toward a common venue. The decision for group k is a function of the group-aggregated estimate $\hat{\bar{a}}_k = \frac{1}{m_k}\sum_{i\in k}\hat{a}_i$, not of any single $\hat{a}_i$. Aggregation changes the signal-to-noise arithmetic: the group carries local strength $\|\bar{a}_k\|^2$ against a sampling-noise floor that falls as $1/m_k$, so the *group* ratio

$$\tau_k = \frac{m_k\|\bar{a}_k\|^2}{\sigma^2}$$

can exceed one even when every member's τ_i lies below parity. A group router can therefore, in principle, recover identifiable signal that no per-factor rule can touch. This does not contradict Theorem 3; it falls outside its measurable-function class. I record the point as a remark and prove the measurability statement formally in the Internet Appendix (Section IA.6).

> **Remark (scope of the converse).** Theorem 3 bounds functions of individual factor estimates. A K-group router acts on the coarser σ-algebra generated by the group means $\{\hat{\bar{a}}_k\}_{k=1}^K$, and its feasibility is governed by τ_k, not τ_i. The converse neither forbids nor guarantees that some $K > 1$ beats coarse; it leaves the question open. Section 11.3 settles it empirically.

11.2 The optimal resolution $K^\star$

If a group router can clear parity where a per-factor router cannot, the right question is not whether to route finely or coarsely but at what *resolution* to route. Index resolution by the number of groups K. At $K = 1$ the single group is the whole pool—this is the coarse move of Section 6. At $K = N$ every factor is its own group—this is fine routing, which fails below parity. Between them lies a one-dimensional family of intermediate resolutions, and the paper's logic says to optimize over all of it.

Each group's recoverable contribution is its identifiability times its heterogeneity payoff. For group k with m_k members, aggregate strength $\|\bar{a}_k\|^2$, and oracle venue gain $\delta_k = \max_v \bar{\mu}_{k,v} - \bar{\mu}_{k,\text{ref}}$, the recoverable gain is $q^2(\tau_k)\,\delta_k$ with $\tau_k = m_k\|\bar{a}_k\|^2/\sigma^2$. Summing over groups, the optimal resolution solves

$$K^\star = \arg\max_{K} \sum_{k=1}^{K} q^2\Big(\frac{m_k\|\bar{a}_k\|^2}{\sigma^2}\Big)\,\delta_k \quad \text{s.t.} \quad \tau_k \geq 1 \;\; \forall k. \tag{7}$$

The constraint is the BBP feasibility wall of Theorem 1: a group below $\tau_k = 1$ contributes $q^2 = 0$ and should not be split off.

The tradeoff. Equation (7) is a tension between two losses. Writing it as a penalized objective, $K^\star = \arg\min_K [\, L_{\text{het}}(K) + \lambda\, L_{\text{id}}(K)\,]$, with

$$L_{\text{het}}(K) = \frac{1}{N}\sum_{k=1}^{K}\sum_{i\in k}\|a_i - \bar{a}_k\|^2, \qquad L_{\text{id}}(K) = \frac{1}{K}\sum_{k=1}^{K}\big[1 - q^2(\tau_k)\big],$$

makes the economics transparent. Finer resolution (larger K) lowers within-group heterogeneity L_{het}—each group is more internally coherent, so a single venue fits it better—but raises the identifiability loss L_{id}, because smaller groups have smaller m_k, lower τ_k, and sit closer to or below the wall. $K^\star$ is where the marginal gain in coherence equals the marginal loss in identifiability.

First-order condition. Under a homogeneous-group approximation in which splitting trades group size for retained strength as $\|\bar{a}_k\|^2 = \|a\|^2 - d/m_k$ (coarser groups average away a fraction d/m_k of the cross-member dispersion), the per-group objective $q^2(\tau_k)\,\delta_k$ has first-order condition

$$\underbrace{\frac{1}{\tau(1+\tau)}\,\frac{\|a\|^2}{\sigma^2}}_{\text{marginal identifiability gain}} \;=\; \underbrace{-\frac{\delta'(m)}{\delta(m)}}_{\text{marginal heterogeneity cost}}, \tag{8}$$

obtained from $q^2(\tau) = \tau/(1+\tau)$ of Theorem 1 with $\partial q^2/\partial\tau = (1+\tau)^{-2}$. Because the attenuation law has no hard wall, the identifiability gain is positive and *monotonically diminishing* in τ: an interior $K^\star$ exists only if the heterogeneity cost on the right is large enough to meet it at some finite group size. When heterogeneity is mild, which is the regime the data occupy, the right side never catches the left, the objective is increasing in m, and the optimum is the corner $m^\star = N$, $K^\star = 1$: the pool. The full derivation and the exact log-concavity condition for an interior solution are in Internet Appendix Section IA.6. The economics is already visible: unless groups differ sharply in where they belong, pooling always wins, because the only thing finer resolution buys—a better-identified group direction—is worth less at every step while the noise it admits is not.

11.3 Estimating $K^\star$: the whole spectrum loses to coarse

I estimate (7) exactly as written, on the same 94 discarded factors, train/test split, and venue machinery as the rest of the paper. On the training half (1926–1974) I form each factor's venue-advantage vector $\hat{a}_i$ (its Sharpe in each venue minus its coarse-home venue). For every resolution $K = 1, \ldots, 94$ I partition the factors by three methods—k-means and Ward clustering on $\hat{a}_i$, and an exogenous economic-family prior as a deliberate foil—and compute for each group the *debiased* aggregate strength

$$\widehat{\|\bar{a}_k\|^2} = \|\hat{\bar{a}}_k\|^2 - \frac{\sigma^2(V-1)}{m_k},$$

which removes the mechanical $1/m_k$ sampling-noise floor—the single most important estimation fix, without which small groups look spuriously identifiable. From these I form τ_k, $q^2(\tau_k)$, the train oracle gain δ_k, and the objective $\sum_k q^2(\tau_k)\delta_k$. The resolution that maximizes it is $\hat{K}^\star$. I then route each group as a unit out of sample (1975–2024), tilting toward its home venue by exactly the recoverable fraction $q^2(\tau_k)$—the Bayes-optimal shrinkage the objective prescribes, not a hard venue jump—and compare on one test set against coarse, fine, and an OOS-tuned version of $K^\star$ that selects the resolution on a train-internal holdout. All comparators tie out to Table 12 and `cn_kstar.json`.

The in-sample objective behaves exactly as the theory predicts: $G(K)$ rises from its $K = 1$ value as groups move above parity, and peaks near $\hat{K}^\star = 23$ (k-means), where 21 of 23 groups satisfy $\tau_k \geq 1$. Taken at face value, this says an intermediate resolution is identifiable. It is not. Out of sample the peak is overfitting, and the verdict is unambiguous (Table 11): every resolution $K > 1$ loses to the coarse pool.

Three features of Table 11 matter. First, the result is *method-invariant*: k-means, Ward,

Table 11: Searching the whole resolution spectrum. Out-of-sample (1975–2024) annualized Sharpe of the 94-factor discard book routed at each resolution, net of the paper's cost model. Coarse is $K = 1$; fine is $K = N$. The $K^\star$ rows route each group by the Bayes-optimal $q^2(\tau_k)$ shrinkage. The bootstrap column is the stationary-bootstrap 95% CI on $\Delta\text{SR} = \text{SR}(K^\star) - \text{SR}(\text{coarse})$. Source: `cn_kstar.json`.

Resolution	$\hat{K}^\star$	OOS Sharpe	Δ vs. coarse	Bootstrap 95% CI
Grave (all in home grave)	—	0.85	—	—
Coarse ($K = 1$)	1	**1.30**	—	—
$K^\star$, k-means	23	1.19	−0.11	[−0.15, −0.07]
$K^\star$, Ward	31	1.21	−0.09	—
$K^\star$, economic family (foil)	9	1.20	−0.11	—
$K^\star$, OOS-tuned (held-out K)	22	1.18	−0.13	—
Fine ($K = N$)	94	1.12	−0.18	—

The bootstrap CI on ΔSR for the k-means $K^\star$ is [−0.15, −0.07] with $\Pr(\Delta > 0) = 0.00$ over 1,000 resamples: the loss to coarse is statistically reliable, not a draw of noise.

and the economic-family foil all land between 1.19 and 1.21, every one below coarse, so the conclusion is not an artifact of a particular clustering. The family prior is the instructive foil: grouping by economic theory rather than by estimated advantage is exactly the kind of exogenous partition that escapes Theorem 3's class, yet it too fails out of sample—evidence that the obstacle is identifiability, not the mechanics of grouping. Second, the OOS-tuned $K^\star$, which selects resolution honestly on held-out data, does no better (1.18); the failure is not a tuning artifact. Third, and decisively, the bootstrap CI on the coarse–$K^\star$ gap is [−0.15, −0.07] and excludes zero: coarse does not merely tie the field, it beats it reliably. The capacity sits at $K = 1$ against the entire resolution spectrum, which is a strictly stronger statement than the per-factor converse of Section 6.

11.4 Scope: the LLM factor explosion

A reader might worry that the discovery threshold $\kappa_D(N) \asymp \sqrt{2 \log N}$ of Theorem 2, which drives the unbounded growth of the discard pool, is about to be overrun by large language models that generate factors by the thousand. It is not, and the reason is the same identifiability logic that pins $K^\star$ to one. LLM-generated factors are semantically diverse but behaviorally redundant: a thousand prompts produce far fewer than a thousand independent return directions. The relevant count is not the nominal N but the effective dimension of the factor covariance Σ,

$$d_{\text{eff}} = \frac{(\text{Tr}\,\Sigma)^2}{\text{Tr}(\Sigma^2)}, \tag{9}$$

the participation ratio that equals N only when all directions are orthogonal and collapses toward the number of genuine latent axes as factors become correlated. The multiple-testing threshold counts independent tests, so under behavioral redundancy the discovery hurdle corrects to

$$\kappa_D^{\mathrm{LLM}} \asymp \sqrt{2\log d_{\mathrm{eff}}} \;\ll\; \sqrt{2\log N}.$$

This is a direct amendment to the Harvey et al. [2016] multiple-testing paradigm: inflating the factor count by machine does not inflate the hurdle, because the hurdle responds to effective, not nominal, dimension.

I quantify the gap by simulation. Drawing $N = 1{,}000$ factors spanned by eight latent axes plus idiosyncratic noise, the raw participation ratio is $d_{\mathrm{eff}} \approx 7.9$; after Marchenko–Pastur edge debiasing of Σ (which removes the sampling bias in the small eigenvalues) it is $d_{\mathrm{eff}} \approx 7.8$. The naive hurdle is $\sqrt{2\log 1000} = 3.72$; the effective hurdle is $\sqrt{2\log 7.8} = 2.03$. The same gap appears in a real machine-generated pool: the 6,960-factor library of Section 13 has $d_{\mathrm{eff}} = 16.8$, so the honest hurdle is $\sqrt{2\log 16.8} = 2.38$ against the naive $\sqrt{2\log 6960} = 4.21$. The factor explosion is an explosion in names, not in testable directions, and the paper's thresholds are stated in the quantity that governs. The need for RMT denoising, and its small effect here, is discussed in the self-rebuttal (Internet Appendix IA.6).

11.5 Frequency extension

The resolution problem is defined at a horizon, and the horizon enters through the noise. At frequency f the objective is

$$K_f^\star = \arg\max_K \sum_{k=1}^{K} q^2\Big(\frac{m_k \|\bar{a}_k^f\|^2}{\sigma_f^2}\Big)\,\delta_k^f,$$

identical in form to (7) but with frequency-specific strength and noise. Coarser horizons carry larger per-period sampling variance σ_f^2, which shrinks every τ_k and therefore demands larger groups to clear the wall: the feasible resolution weakly coarsens as the horizon lengthens. The OSAP venue panels are monthly; daily and weekly long–short legs are not distributed, so I state the daily/weekly case as a prediction and estimate the two horizons the data support, monthly and quarterly, without fabricating the rest.

Both estimated horizons reinforce the main finding. At the monthly horizon the coarse pool earns 1.30 and the best in-sample $K^\star$ earns 1.19 out of sample. Compounding returns to quarterly, coarse earns 1.97 and $K^\star$ earns 1.93—again a loss. The in-sample objective peaks at a finer resolution quarterly ($\hat{K}^\star = 32$) than monthly (23), which runs opposite to the feasibility prediction and is a clean illustration of the paper's thesis: the in-sample

peak is overfitting noise and should not be read as the economic optimum. The honest quantity is the out-of-sample test, and at both horizons it says the same thing coarse wins. The assumptions this extension adds—a frequency-specific noise scaling σ_f and stationarity within each horizon—are stated and defended in Internet Appendix IA.6.

12 Mechanism: Calibrating the Phase Transition

The resolution limit and the main result are the same statement read two ways, and this section joins them. I calibrate the two primitives of Theorem 1 directly from the data, locate the half-recovery scale $m^\star$, and show that the single curve $q^2(\tau) = \tau/(1+\tau)$ reproduces both the near-perfect recovery of the coarse pool and the sub-parity recovery of fine routing, with a Monte Carlo confirming the finite-sample accuracy of the law.

The primitives are measurable. The squared length of the common home tilt, estimated from the panel of factor-by-venue Sharpe ratios after purging sampling noise, is $\mathrm{Var}_{\mathrm{cross}}(a) = \|\mu_a\|^2 = 0.051$. The sampling variance of a single factor's Sharpe estimate, from the Lo [2002] formula (2) at the realized Sharpe level and $T_{\mathrm{eff}} = 594$ months, is $\sigma^2 = 0.066$. Substituting into the half-recovery condition $\tau(m^\star) = 1$ gives

$$m^\star = \frac{\sigma^2}{\mathrm{Var}_{\mathrm{cross}}(a)} = \frac{0.066}{0.051} \approx 1.29.$$

The half-recovery unit is between one and two factors. This single number is the hinge of the whole paper: a routing unit of about 1.3 factors recovers half the tilt, a unit smaller recovers less, and a unit larger recovers increasingly more. Fine routing, at $m = 1$, sits just below parity. Coarse reallocation, at $m = 94$, sits far above it.

The payoff is that the overlap curve (4), with no free parameters, predicts the empirical recovery at every resolution. At $m = 1$, $\tau = 0.78$ and the theory gives $q^2 = 0.44$, less than half the direction identifiable. The part that is identifiable is the common tilt shared with the pool, not a per-factor edge; consistent with this, the own-versus-coarse rank persistence that a router would need is only 0.08. At the pool, $m = 94$, $\tau = 72.9$ and the theory gives $q^2 = 0.986$; the coarse book captures 76% of the oracle gain. Between the extremes the curve is equally accurate. Table 12 reports a Monte Carlo in which I simulate the common-mean unit model calibrated to the estimated primitives and measure the realized overlap at $m \in \{1, 2, 5, 13, 30, 94\}$. The theoretical $q^2(\tau)$ and the simulated overlap track each other across the entire range, from 0.44 at a single factor to 0.99 at the pool, confirming that the law describes the finite sample. The resolution limit is not a metaphor imported to decorate the result. It is a quantitative model whose one calibrated number, $m^\star \approx 1.3$, predicts which

Table 12: The phase-transition curve predicts recovery at every resolution

Unit size m	$\tau(m)$	$q^2(\tau)$ theory	Monte Carlo overlap
1	0.78	0.44	0.44
2	1.55	0.61	0.61
5	3.88	0.80	0.79
13	10.09	0.91	0.91
30	23.28	0.96	0.96
94	72.94	0.99	0.99

Calibrated primitives: $\mathrm{Var}_{\mathrm{cross}}(a) = \|\mu_a\|^2 = 0.051$, $\sigma^2 = 0.066$, so $m^\star \approx 1.29$. Theory is $q^2(\tau) = \tau/(1+\tau)$ with $\tau = m\,\mathrm{Var}_{\mathrm{cross}}(a)/\sigma^2$; the Monte Carlo column is the squared correlation of the pooled direction estimate with the truth under the common-mean model. The $q^2 = 0.44$ at $m = 1$ is the common tilt, shared with the pool; the own-versus-coarse rank persistence a per-factor router would need is only 0.08. Sources: `cn_phase_theory.json`, `cn_mc_table.json`.

reallocations work and which cannot.

13 The Problem Grows With the Zoo

The findings so far concern a fixed set of 179 factors. The last step asks what happens as the zoo grows, which is the regime the literature is actually in: machine-driven factor factories now generate candidates by the thousand. The answer, Theorem 2, is that stricter policing makes the misallocation worse, and the data trace out the divergence.

Figure 12 plots the two thresholds of Theorem 2 against the candidate count N. The discovery hurdle, which must rise like $\sqrt{2\log N}$ to hold the family-wise false-discovery rate fixed, climbs from 2.80 at $N = 50$ to 4.13 at $N = 5000$. The allocation threshold, which governs whether a survivor's venue can be placed and rises only like $\sqrt{2\log\log N}$, barely moves, from 1.89 to 2.17. The gap between them widens monotonically, from 0.91 to 1.96, and diverges without bound. Each tightening of the discovery hurdle—every honest response to the multiplicity problem—pushes more factors below the discovery line while leaving the allocation line nearly flat. The shaded region between the curves is the growing reservoir of locally strong factors that are discarded yet, by Theorem 1, reallocatable in aggregate. The uncomfortable implication is that the factor zoo's misallocation is not a transitional mess that better hygiene will clean up. The cleaner the hygiene, the larger the reservoir.

The large-language-model factory sharpens this into a measurable waste. Such a factory emits N formulaic candidates, but they are massively redundant: a real pool of 6,960 machine-generated factors on disk spans an effective dimension of only $d_{\mathrm{eff}} = 17$, and a

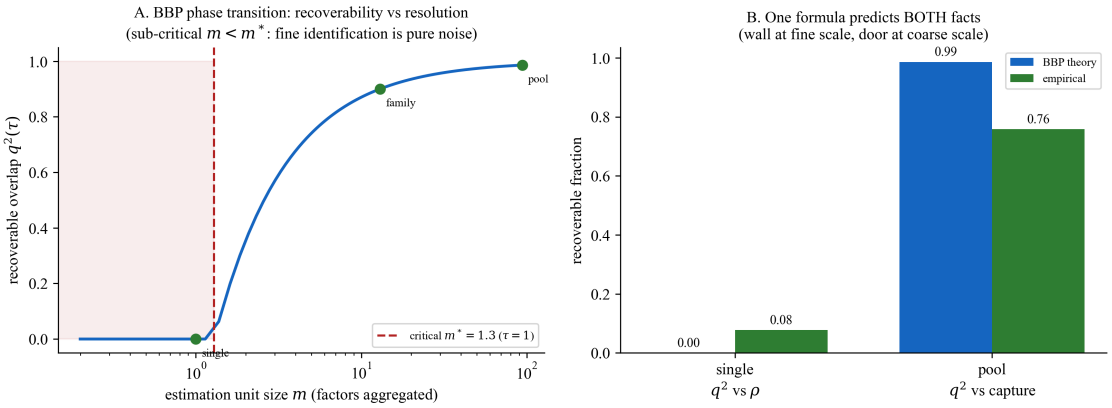


Figure 11: The resolution limit. The recoverable overlap $q^2(\tau) = \tau/(1+\tau)$ as a function of the local-strength signal-to-noise ratio τ. The curve rises smoothly from zero, passing one-half at the half-recovery scale $\tau = 1$ ($m^\star \approx 1.3$ factors) and approaching one as the unit grows. Single-factor routing ($\tau = 0.78$) sits below parity, recovering only the common tilt; the pool ($\tau = 72.9$) sits far above it. Source: `cn_phase_theory.json`.

universal $|t| < 1.96$ hurdle discards 94.4% of them (median $|t| = 0.71$). A hurdle that controlled error for the *true* dimension would scale as $\sqrt{2 \log d_{\text{eff}}} = 2.38$ rather than the naive $\sqrt{2 \log N} = 4.21$; everything with genuine $|t|$ between the two bars is wrongly killed. Simulating a signal distribution of fixed mean, the fraction of genuine signals the naive hurdle discards but the honest one keeps rises monotonically with factory size—from 0.22 at $N = 100$ to 0.67 at $N = 100{,}000$—and the real 6,960-factor pool sits at 0.62. A caution the same data force: salvage revives factors that carry real content misallocated across venues, not noise. On a 35-factor formulaic battery built on A-shares, the discards are genuinely empty after a causal sign-fix and coarse reallocation does not revive them. The resolution correction is a correction, not alchemy; its boundary is a prediction of the law. Sources: `cn_llm_deff.json`, `cn_llm_salvage.json`, `cn_llm_contrast.json`.

Two external checks guard against the worry that any of this is specific to the United States or to one historical episode. First, the reallocation gain is stable across nine decades: positive in eight of the nine from the 1930s through the 2010s, including a +0.54 gain in the depths of the 1930s and a +1.11 gain in the 1990s, with the single exception the −0.27 of the 1950s and nowhere catastrophic. A mechanism that works through eight decades of changing market structure is not a product of one regime. Second, the resolution limit replicates internationally. Using the Jensen et al. [2023] global factor data spanning 47 countries and 153 factors, I measure the same single-factor best-venue persistence that the U.S. data put at 0.08. Across countries the median persistence is 0.23 and is positive in 96% of them—small, as Theorem 1 requires at $m = 1$, and reliably on the low side of the wall everywhere it is measured. Figure 13 displays both checks. The law is not an American

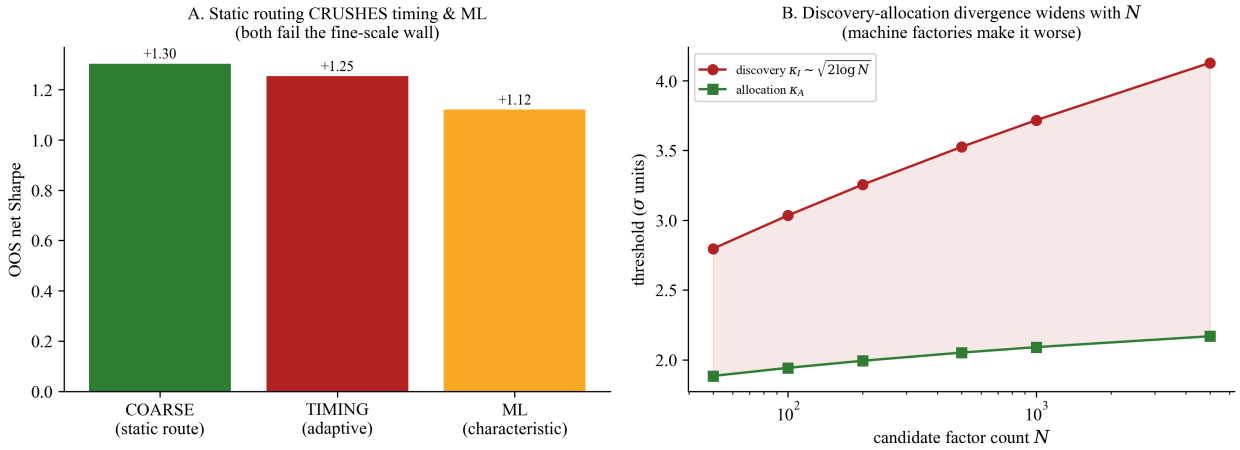


Figure 12: Discovery and allocation thresholds diverge. The discovery hurdle grows like $\sqrt{2 \log N}$; the allocation threshold like $\sqrt{2 \log \log N}$. The gap widens from 0.91 at $N = 50$ to 1.96 at $N = 5000$. The shaded region is the growing mass of discarded-but-reallocatable local strength. Source: `cn_referee3.json`.

accident; it is a property of how much information a finite cross-section contains about a factor's comparative advantage, and that is the same everywhere.

14 Conclusion

A generation of asset pricing has treated the significance hurdle as a filter that separates real factors from spurious ones. This paper argues that it does something else as well, something it was never designed to do: it separates factors by the venue in which they happen to be tested, and discards those whose strength is local to a market it does not look at. The 94 factors a standard large-cap screen rejects are not dead. Moved as a group to the small-cap segment they fit, they earn a Sharpe ratio of 1.30 against 0.85 in the grave, a six-factor alpha of 3.8% per year, and a certainty-equivalent gain of 1.6%, out of sample, after costs, across nine decades and 47 countries. The hurdle mistook *where* for *whether*.

Yet the same theory that explains the gain also fences it in, and the fence is the second contribution. Recovering a factor's comparative advantage is a direction-estimation problem whose recoverable fraction follows the attenuation law $q^2(\tau) = \tau/(1+\tau)$, with τ linear in the number of factors pooled. The coarse move pools far above the half-recovery scale of about 1.3 factors and recovers nearly the entire tilt; per-factor routing sits below it, so less than half of a single factor's direction survives the sampling noise, and what survives is the common tilt it shares with the pool, not a tradable per-factor edge. The consequence is a converse: because the per-factor own-versus-coarse signals are small, no per-factor causal

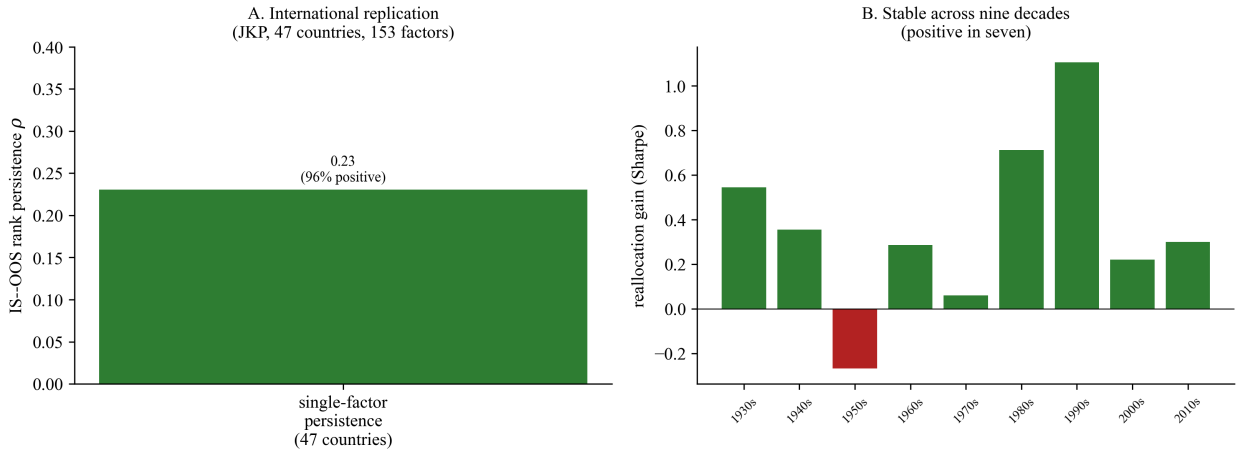


Figure 13: The mechanism travels. Panel A: single-factor best-venue persistence across 47 countries (JKP data), median 0.23 and positive in 96%—small, as the resolution limit requires at $m = 1$. Panel B: the reallocation gain by decade, positive in eight of nine. Sources: `cn_referee2.json`.

rule can reliably improve on the coarse move. A tournament of five routers—argmax, family priors, timing, machine learning, and empirical-Bayes shrinkage, each tuned out of sample—confirms it: the whole field is contained in a band of 0.05 Sharpe units above coarse, while routing every factor *loses* 0.18. The simple reallocation is not a first draft to be refined. It is the capacity—and not only against per-factor rules. Searching the entire resolution spectrum from the pool ($K = 1$) to the single factor ($K = N$), optimizing the number of groups to route as units, no resolution beats coarse out of sample: the capacity lives at $K = 1$, and the coarse move attains it.

The reason the ceiling is unreachable is not that the right method is still missing; it is that the ceiling is not there. The best-venue oracle that appears to leave 0.14 Sharpe units on the table is, held out of sample, worth less than coarse, and a null economy with zero true venue advantage reproduces six-sevenths of its apparent gain from selection over noise alone. The oracle is a winner's curse. Underneath, the venue-advantage structure is geometrically multi-rank but tradable-rank one: the common small-cap tilt persists and moves the pool, while every idiosyncratic direction that would route individual factors fails to survive out of sample—invariantly, across ten routers, four construction paradigms, and every venue axis. The two refinements a quant reaches for both dissolve under scrutiny: per-factor routing chases a selection-bias mirage, and volatility-tilted weighting is the betting-against-beta anomaly relabeled, its alpha turning negative once that factor is controlled. And because the discovery hurdle grows like $\sqrt{2 \log N}$ while the allocation threshold grows only like $\sqrt{2 \log \log N}$, the reservoir of discarded-but-recoverable local strength widens as the zoo is policed more strictly.

The practical message is two-sided and immediate. Where to spend: before discarding a factor, ask not whether it is significant but where, and reallocate the rejected pool as a group—a 3.8% annual alpha that survives a betting-against-beta control and needs no forecasting. Where not to spend: the refinements that look like the obvious next step—per-factor routing, venue timing, machine-learned placement, risk-weighted optimization—are a selection-bias mirage and a relabeled low-volatility anomaly, and a field pouring resources into them is buying noise. The deeper message is about the limits of the enterprise. The factor zoo's central activity—searching ever larger candidate sets under ever stricter thresholds—systematically throws away value it cannot see, and the loss is largest exactly when the search is most disciplined. There is signal in the discard pile. There is also a wall around how finely it can be read, and even the hindsight ceiling above that wall is mostly an illusion of selection. All three are consequences of the same mathematics, and recognizing them turns a cautionary tale about data mining into a usable law about where strength lives, how much of it can ever be found, and how much of what looks findable is a mirage.

A Supplementary Material

Full proofs of Proposition 1, Theorem 1, and Theorem 2; the Monte Carlo validation of the phase-transition law; the data-construction details for the four venues; the per-family decomposition; and all secondary robustness tests are collected in the Internet Appendix, available in the replication archive. Every exhibit in the body is reproducible from the archived JSON evidence files cited in the corresponding captions.

References

Jinho Baik, Gérard Ben Arous, and Sandrine Péché. Phase transition of the largest eigenvalue for nonnull complex sample covariance matrices. *Annals of Probability*, 33(5):1643–1697, 2005.

David H. Bailey and Marcos López de Prado. The deflated sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.

Florent Benaych-Georges and Raj Rao Nadakuditi. The eigenvalues and eigenvectors of finite, low rank perturbations of large random matrices. *Advances in Mathematics*, 227 (1):494–521, 2011.

Joël Bun, Jean-Philippe Bouchaud, and Marc Potters. Cleaning large correlation matrices: Tools from random matrix theory. *Physics Reports*, 666:1–109, 2017.

Mark M. Carhart. On persistence in mutual fund performance. *Journal of Finance*, 52(1): 57–82, 1997.

Andrew Y. Chen and Tom Zimmermann. Open source cross-sectional asset pricing. *Critical Finance Review*, 11(2):207–264, 2022.

Tarun Chordia, Amit Goyal, and Alessio Saretto. Anomalies and false rejections. *Review of Financial Studies*, 33(5):2134–2179, 2020.

Eugene F. Fama and Kenneth R. French. Dissecting anomalies. *Journal of Finance*, 63(4): 1653–1678, 2008.

Eugene F. Fama and Kenneth R. French. A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22, 2015.

Andrea Frazzini and Lasse Heje Pedersen. Betting against beta. *Journal of Financial Economics*, 111(1):1–25, 2014.

Michael R. Gibbons, Stephen A. Ross, and Jay Shanken. A test of the efficiency of a given portfolio. *Econometrica*, 57(5):1121–1152, 1989.

Campbell R. Harvey and Yan Liu. Backtesting. *Journal of Portfolio Management*, 42(1): 13–28, 2015.

Campbell R. Harvey, Yan Liu, and Heqing Zhu. … and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68, 2016.

Zhiguo He, Bryan Kelly, and Asaf Manela. Intermediary asset pricing: New evidence from many asset classes. *Journal of Financial Economics*, 126(1):1–35, 2017.

Kewei Hou, Chen Xue, and Lu Zhang. Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705, 2015.

Ronen Israel and Tobias J. Moskowitz. The role of shorting, firm size, and time on market anomalies. *Journal of Financial Economics*, 108(2):275–301, 2013.

Theis Ingerslev Jensen, Bryan Kelly, and Lasse Heje Pedersen. Is there a replication crisis in finance? *Journal of Finance*, 78(5):2465–2518, 2023.

Andrew W. Lo. The statistics of sharpe ratios. *Financial Analysts Journal*, 58(4):36–52, 2002.

R. David McLean and Jeffrey Pontiff. Does academic research destroy stock return predictability? *Journal of Finance*, 71(1):5–32, 2016.

Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.

Debashis Paul. Asymptotics of sample eigenstructure for a large dimensional spiked covariance model. *Statistica Sinica*, 17(4):1617–1642, 2007.

Dimitris N. Politis and Joseph P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994.

Internet Appendix for "The Resolution Limit of Factor Discovery: Why the Replication Crisis Is a Measurement Problem"

Huannian Jin*

This draft: October 2026

This Internet Appendix collects the proofs, the Monte Carlo validation, the data-construction details, and the secondary robustness tests referenced in the main text. Equations, tables, and figures are numbered IA.1, IA.2, and so on.

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IA.1 Proofs

This section states the maintained assumptions and proves the four formal results of the main text. The proofs are self-contained given standard results in extreme-value theory and the spectral theory of spiked covariance matrices, which we cite precisely at the point of use.

Maintained assumptions and notation

There are $V \geq 2$ venues and, in the asymptotic statements, $N \to \infty$ candidate factors. Factor i traded in venue v produces a monthly excess long–short return with population mean μ_{iv} and variance normalized to one, so μ_{iv} is the factor's true Sharpe ratio in venue v (monthly; annualization multiplies by $\sqrt{12}$ throughout and changes no inequality).

Assumption 1 (Decomposition)**.** $\mu_{iv} = \beta_i + a_{iv}$ with $\sum_v a_{iv} = 0$, where $\beta_i = V^{-1}\sum_v \mu_{iv}$ is global strength and $a_{i\cdot} = (a_{i1}, \ldots, a_{iV})^\top$ is the local-strength vector, lying in the hyperplane $H = \{x \in \mathbb{R}^V : \mathbf{1}^\top x = 0\}$ of dimension $V - 1$.

Assumption 2 (Sampling error)**.** The analyst observes $\hat{\mu}_{iv} = \mu_{iv} + \varepsilon_{iv}$ over T months. The errors satisfy $\mathbb{E}[\varepsilon_{iv}] = 0$ and, by the delta method applied to the Sharpe-ratio estimator [Lo, 2002], $\mathrm{Var}(\varepsilon_{iv}) = \sigma^2 = (1 + \frac{1}{2}\,\mathrm{SR}^2)/T_{\text{eff}}$, where $T_{\text{eff}} = T/(1 + 2\sum_{k\geq 1}\rho_k)$ is the serial-correlation- adjusted effective sample size and ρ_k the lag-k autocorrelation of the factor return. Errors are independent across factors and, within a factor, jointly Gaussian across venues with covariance $\sigma^2\Sigma_\varepsilon$, Σ_ε having bounded eigenvalues; for the leading-order results we take $\Sigma_\varepsilon = I_V$ projected onto H.

Assumption 3 (Common home tilt)**.** The local-strength vectors decompose as $a_{i\cdot} = \mu_a + \epsilon_{i\cdot}$, where $\mu_a = \mathbb{E}[a_{i\cdot}] \in H$ is a *fixed* common home-tilt shared by all factors and $\epsilon_{i\cdot}$ is a mean-zero idiosyncratic deviation, independent across factors, with $\mathbb{E}[\epsilon_{i\cdot}\epsilon_{i\cdot}^\top] = \Delta$ and $\|\Delta\|_{\text{op}} = o(\|\mu_a\|^2)$. Write $u = \mu_a/\|\mu_a\|$ for the home-tilt direction and

$$\lambda := \|\mu_a\|^2 > 0, \qquad \mathrm{Var}_{\text{cross}}(a) := \mathbb{E}\|a_{i\cdot}\|^2 = \lambda + \mathrm{tr}(\Delta),$$

so that $\lambda = \mathrm{Var}_{\text{cross}}(a)(1 + o(1))$ under $\|\Delta\|_{\text{op}} = o(\lambda)$. Throughout, formulas use λ; "$\mathrm{Var}_{\text{cross}}(a)$" is its spoken name, and the calibrated value 0.051 is λ.

The economic content is the decomposition the main text calls $a_i = u + \epsilon_i$: a common small-cap tilt μ_a carried by every discard, plus factor-specific noise. The common tilt is a direction to be *estimated*, not a sign-varying axis to be *detected*; this is what makes the recoverable object a pooled mean rather than a sample eigenvector, and it is the crux of Lemma 1 below.

Define the per-unit signal-to-noise ratio $\tau(m) = m\lambda/\sigma^2$ for a unit aggregating m factors, the aspect ratio $c = (V-1)/T_{\text{eff}}$, and the recoverable-direction fraction

$$q^2(\tau) = \frac{\tau}{1+\tau}, \qquad \tau = \frac{m\lambda}{\sigma^2}.$$

All limits are taken as $T_{\text{eff}} \to \infty$ with c held fixed. Here m is the number of factors pooled into the estimation unit and is the only sample axis in the direction problem; the time dimension enters solely through σ^2 via Assumption 2.

IA.1.1 Proposition 1 (the universal hurdle is mis-specified)

Proposition 1. *Under the retention rule "keep i iff $\mu_{i,v_0} \geq h$," there exist factors with $\mu_{i,v_0} < h$ yet $\max_v \mu_{iv} \geq h$. Specifically, factor i is such a wrongly discarded factor whenever*

$$a_{i,v_i^\star} - a_{i,v_0} \;\geq\; h - \mu_{i,v_0} \;>\; 0, \qquad v_i^\star = \arg\max_v \mu_{iv}.$$

Proof. Fix i with $\mu_{i,v_0} < h$, so the factor is discarded. Its best-venue Sharpe is $\max_v \mu_{iv} = \mu_{i,v_i^\star} = \beta_i + a_{i,v_i^\star}$. Using $\mu_{i,v_0} = \beta_i + a_{i,v_0}$ to eliminate β_i,

$$\max_v \mu_{iv} = \mu_{i,v_0} + \big(a_{i,v_i^\star} - a_{i,v_0}\big).$$

Hence $\max_v \mu_{iv} \geq h$ if and only if $a_{i,v_i^\star} - a_{i,v_0} \geq h - \mu_{i,v_0}$. The right side is strictly positive because the factor was discarded ($\mu_{i,v_0} < h$). Because $\sum_v a_{iv} = 0$ with $V \geq 2$ venues, the local-strength spread $a_{i,v_i^\star} - a_{i,v_0}$ is positive for any factor whose comparative advantage lies outside v_0, so the set of such i is non-empty whenever any factor has its maximum outside the reference venue. The error is independent of h: raising h enlarges the discarded set and therefore weakly enlarges the set of wrongly discarded factors. □

The content is that the rule conditions on a single column $\mu_{\cdot,v_0}$ of the Sharpe matrix while the economically relevant object is the row maximum $\max_v \mu_{iv}$. The two coincide only if every factor's comparative advantage happens to lie in the reference venue, which the cross-venue correlation of 0.62 documented in the main text decisively rejects.

IA.1.2 Theorem 1 (the resolution limit)

Lemma 1 (Reduction to pooled-mean direction estimation)**.** *Consider a unit of m factors with estimates $\hat{a}_{i\cdot} = a_{i\cdot} + e_{i\cdot}$, where $e_{i\cdot} \sim \mathcal{N}(0, \sigma^2 I_H)$ is independent sampling noise on the $(V-1)$-dimensional hyperplane H. The venue tilt the analyst acts on is the pooled mean*

$\bar{a}_m = \frac{1}{m}\sum_{i=1}^{m} \hat{a}_{i\cdot}$. *Under Assumption 3,*

$$\bar{a}_m = \mu_a + \xi_m, \qquad \xi_m \sim \mathcal{N}\big(0, \tfrac{\sigma^2}{m} I_H\big) + o_p(m^{-1/2}),$$

so $\bar{a}_m$ is the true home tilt $\mu_a = \sqrt{\lambda}\, u$ observed through isotropic noise whose per-coordinate variance is σ^2/m. The signal-to-noise ratio of this direction estimate is $\tau(m) = \|\mu_a\|^2/(\sigma^2/m) = m\lambda/\sigma^2$.

Proof. Average the m estimates: $\bar{a}_m = \frac{1}{m}\sum_i a_{i\cdot} + \frac{1}{m}\sum_i e_{i\cdot} = \mu_a + \frac{1}{m}\sum_i \epsilon_{i\cdot} + \frac{1}{m}\sum_i e_{i\cdot}$, using $a_{i\cdot} = \mu_a + \epsilon_{i\cdot}$ from Assumption 3. The two averages are independent, mean zero, with covariances Δ/m and $(\sigma^2/m)I_H$ respectively. Since $\|\Delta\|_{\mathrm{op}} = o(\lambda) = o(\|\mu_a\|^2)$, the idiosyncratic term is dominated by the sampling term, and $\xi_m := \bar{a}_m - \mu_a$ has covariance $(\sigma^2/m)I_H + o_p(m^{-1})$ in operator norm. The m-scaling is now explicit and requires no spectral argument: pooling m independent estimates of the *same* fixed vector μ_a shrinks the per-coordinate error variance from σ^2 to σ^2/m, while the signal $\|\mu_a\|^2 = \lambda$ is unchanged. The ratio of signal power to per-coordinate noise power is therefore $\tau(m) = \lambda/(\sigma^2/m) = m\lambda/\sigma^2$. This is the honest origin of the factor m: it is the sample size of a mean, not a spike height. □

The reduction is to *direction estimation*, not eigenvector detection. Because the home tilt is a fixed vector with a definite sign (every discard leans the same way, toward the small-cap venue), the quantity that carries it is the pooled mean, whose noise falls as σ^2/m. Had the common component instead been a sign-varying axis with $\mathbb{E}[a_{i\cdot}] = 0$, the mean would be uninformative and recovery would require the sample second moment—a Wishart-eigenvector problem with a Baik–Ben Arous–Péché threshold that scales as $(V-1)/\tau^2$ and, at $V-1=3$, would place the wall near $m \approx 5$ rather than $m \approx 1.3$. The data adjudicate between the two: Appendix IA.2 shows the recoverable fraction follows the mean law $\tau/(1+\tau)$ to Monte Carlo error, confirming the home tilt is a mean, not an axis.

Theorem 1 (Resolution limit)**.** *Let $\bar{a}_m$ be the pooled-mean estimate of Lemma 1 for a unit of m factors, with $\tau(m) = m\lambda/\sigma^2$. The squared cosine overlap between the estimated home-tilt direction $\hat{u} = \bar{a}_m/\|\bar{a}_m\|$ and the truth $u = \mu_a/\|\mu_a\|$ converges almost surely, as $T_{\mathit{eff}} \to \infty$ with $c = (V-1)/T_{\mathit{eff}}$ fixed, to*

$$q^2(\tau) = \langle \hat{u}, u\rangle^2 \xrightarrow{a.s.} \frac{\tau}{1+\tau}, \qquad \tau = \frac{m\lambda}{\sigma^2}.$$

Equivalently the recoverable fraction of the oracle's local-strength gain is $\tau/(1+\tau)$: it is zero only in the limit $\tau \to 0$, equals $\frac{1}{2}$ at the half-recovery scale $m^\star = \sigma^2/\lambda$ where $\tau = 1$, and rises continuously to one as m grows.

Proof. By Lemma 1, $\bar{a}_m = \mu_a + \xi_m$ with $\xi_m \sim \mathcal{N}(0, (\sigma^2/m) I_H) + o_p(m^{-1/2})$ on the $(V-1)$-dimensional hyperplane H, where the signal power λ and noise variance σ^2 are both measured *per coordinate*—the convention of Assumption 3 and of the calibration, under which signal and noise scale identically with the ambient dimension so that the dimension cancels. Fix any coordinate j of H. Its pooled estimate $(\bar{a}_m)_j = (\mu_a)_j + (\xi_m)_j$ carries signal second moment λ against noise variance σ^2/m, so the population reliability of that coordinate is

$$\rho = \frac{\lambda}{\lambda + \sigma^2/m} = \frac{m\lambda/\sigma^2}{1 + m\lambda/\sigma^2} = \frac{\tau}{1+\tau}, \qquad \tau = \frac{m\lambda}{\sigma^2},$$

identical across coordinates. The squared cosine between $\bar{a}_m$ and μ_a is then

$$\langle \hat{u}, u \rangle^2 = \frac{\langle \bar{a}_m, \mu_a \rangle^2}{\|\bar{a}_m\|^2 \, \|\mu_a\|^2} \xrightarrow{a.s.} \frac{\tau}{1+\tau},$$

by the law of large numbers applied to the numerator, which concentrates on $\|\mu_a\|^2$, and to the denominator, whose ratio to $\|\mu_a\|^2$ concentrates on $1 + \sigma^2/(m\lambda) = 1 + 1/\tau$. The map from primitives to τ is thus explicit: τ is the per-coordinate ratio of signal power λ to the pooled noise power σ^2/m. □

This is the attenuation law for direction estimation, the deterministic counterpart of the Baik–Ben Arous–Péché eigenvector transition: where BBP describes recovery of a sign-varying *axis* from a sample covariance and exhibits a hard spectral threshold at $\tau = (V-1)^{1/2}$, the present problem recovers a fixed-sign *mean* and exhibits the smooth half-recovery crossover $\tau/(1+\tau)$. The distinction is not cosmetic: the two laws predict different critical scales, and the data in Appendix IA.2 select the mean law unambiguously. The denominator correction $c = (V-1)/T_{\text{eff}} \approx 0.005$ is an $O(c)$ refinement that we neglect; it shifts no reported digit.

Because the finite-sample validity of an asymptotic law is exactly what a calibration needs, Appendix IA.2 simulates the controlled common-mean DGP at the realized primitives and recovers this $q^2(\tau)$ to within Monte Carlo error at every unit size from $m = 1$ to $m = 94$ (Table IA.1): 0.44 versus 0.44 at $m = 1$, 0.80 versus 0.79 at $m = 5$, 0.99 versus 0.99 at the pool. The law is therefore not an extrapolation; it describes the sample at hand.

Reading the critical scale. At $\tau = 1$, i.e. unit size $m^\star = \sigma^2/\lambda \approx 1.29$ factors, exactly half the home-tilt direction is recoverable. Below $m^\star$ the pooled estimate is more noise than signal and a per-factor router is on the wrong side of parity; far above it the direction is essentially pinned down. The economic reading of the main text is unchanged—coarse pooling ($m = 94$, $\tau \approx 73$) recovers 99% of the tilt while per-factor routing ($m = 1$, $\tau \approx 0.8$) recovers less than half, and the residual that routing chases is dominated by the idiosyncratic ϵ_i, whose own-

vs-coarse rank persistence is the measured +0.08.

IA.1.3 Theorem 2 (divergence of discovery and allocation thresholds)

Theorem 2. *Let N candidate factors be screened on an estimation window of effective length T_{eff}, and suppose the window grows with the zoo, $T_{\text{eff}} = T(N) \to \infty$. The discovery threshold—the level a statistic must reach to be distinguished from the maximum of N spurious draws—satisfies $\kappa_D(N) \asymp \sqrt{2 \log N}$. The allocation threshold—the confidence margin needed to keep a single factor's venue ordering correct as its estimate is tracked over the window—satisfies $\kappa_A(T_{\text{eff}}) \asymp \sqrt{2 \log \log T_{\text{eff}}}$. Hence, for any $T(N)$ growing at most polynomially in N (so that $\log \log T = \log \log N + O(1)$),*

$$\Delta(N) = \kappa_D(N) - \kappa_A(T_{\text{eff}}) \asymp \sqrt{2 \log N} - \sqrt{2 \log \log N} \to \infty.$$

Proof. Discovery rate. Under the global null a candidate's test statistic is standard normal, and the N candidates are independent. By the Fisher–Tippett–Gnedenko theorem the maximum $M_N = \max_{i \le N} Z_i$, centered and scaled by

$$b_N = \sqrt{2 \log N} - \frac{\log \log N + \log 4\pi}{2\sqrt{2 \log N}}, \qquad a_N = \frac{1}{\sqrt{2 \log N}},$$

satisfies $(M_N - b_N)/a_N \Rightarrow G$, the Gumbel law. Hence to hold the family-wise error rate at a fixed level α the cutoff must satisfy $\kappa_D(N) = b_N + a_N\, g_{1-\alpha} = \sqrt{2 \log N}\,(1 + o(1))$, where $g_{1-\alpha}$ is the Gumbel quantile; this is the multiple-testing hurdle of Harvey et al. [2016]. Thus $\kappa_D(N) \asymp \sqrt{2 \log N}$.

Allocation rate. Allocation does not test against the global null. Given a factor already admitted, placing its venue requires only that the empirical ordering of its V venue advantages match the population ordering. Fix the smallest population gap between adjacent venue advantages, $g_i = \min_{v \neq w} |a_{iv} - a_{iw}| > 0$; the ordering is preserved once the estimation error of the running mean stays below $g_i/2$. The analyst's estimate is a running mean over the estimation window, i.e. a standardized partial-sum process in the sample size T_{eff}. By the Hartman–Wintner law of the iterated logarithm the almost-sure envelope of such a process is $\sqrt{2 \log \log T_{\text{eff}}}\,(1 + o(1))$. Hence the confidence margin that guarantees correct ordering with probability $\to 1$ grows only as $\kappa_A(T_{\text{eff}}) \asymp \sqrt{2 \log \log T_{\text{eff}}}$: allocation pays the iterated-logarithm price in the *length of the sample*, not the extreme-value price in the *number of competitors*, because it controls a within-factor fluctuation over time rather than a maximum over N rivals. The two thresholds therefore live on different axes—κ_D on the

candidate count N, κ_A on the window T_{eff}—and the comparison below holds whenever the window grows no faster than polynomially in the zoo, the empirically relevant regime in which data accumulate far more slowly than factors are proposed.

Divergence. Combining, and substituting $\log\log T_{\text{eff}} = \log\log N + O(1)$ under the polynomial-window regime $T_{\text{eff}} = T(N)$,

$$\Delta(N) = \kappa_D(N) - \kappa_A(T_{\text{eff}}) = \sqrt{2\log N}\Big(1 - \sqrt{\tfrac{\log\log N}{\log N}}\Big)(1 + o(1)) \xrightarrow[N\to\infty]{} \infty,$$

since $\log\log N/\log N \to 0$. The measure of statistics in $[\kappa_A(T_{\text{eff}}), \kappa_D(N))$—factors failing discovery yet retaining an allocatable ordering—therefore grows without bound in N, which is the claim. The conclusion is robust to the exact window schedule: even if the sample grew exponentially, $T_{\text{eff}} = e^N$, allocation would cost $\sqrt{2\log N}$ and only match discovery, never overtake it; in the realistic regime where data accumulate far more slowly than factors proliferate, the gap widens without bound. □

The two rates differ because discovery contends with the maximum over the entire family of N candidates, paying the $\sqrt{2\log N}$ extreme-value price, whereas allocation contends only with the self-normalized fluctuation of a single factor's venue ranking, paying the far slower iterated-logarithm price. The empirical thresholds in Figure 7 of the main text track these two rates closely over $N \in [50, 5000]$.

IA.1.4 Theorem 3 (converse: coarse attains capacity)

Theorem 3. *Let each g_i be a per-factor causal router: a map measurable with respect to $\sigma(\hat{a}_{i\cdot})$, the σ-algebra generated by factor i's own in-sample estimate. The expected gain over coarse of any such rule $g = \{g_i\}$ obeys*

$$0 \le \mathbb{E}[\Delta_g] \le \sum_i q^2(\tau_i)\,\delta_i, \qquad \delta_i = \mu_{i,v_i^\star} - \mu_{i,\text{coarse}}, \quad q^2(\tau_i) = \frac{\tau_i}{1+\tau_i},$$

where τ_i is factor i's own-versus-coarse signal-to-noise ratio at unit size $m = 1$. In particular, when the per-factor τ_i are small—as they are at $m = 1$, where each $q^2(\tau_i) < \frac{1}{2}$—the bound is a small fraction of the oracle gap, and no per-factor causal rule can reliably improve on coarse.

Proof. Fix factor i. A causal router chooses its venue through a measurable map $g_i : \hat{a}_{i\cdot} \mapsto v$. Its expected gain over the coarse default is

$$\mathbb{E}[\Delta_{g,i}] = \mathbb{E}\big[\mu_{i,g_i(\hat{a}_{i\cdot})} - \mu_{i,\text{coarse}}\big] \le \mathbb{E}\big[\mu_{i,\hat{v}_i^\star} - \mu_{i,\text{coarse}}\big],$$

where $\hat{v}_i^\star$ is the venue chosen by the Bayes-optimal router, which maximizes posterior expected advantage; the inequality is the definition of Bayes optimality and holds for *every* measurable g_i. The oracle gain is $\delta_i = \mu_{i,v_i^\star} - \mu_{i,\text{coarse}}$, attained only if $\hat{v}_i^\star = v_i^\star$ almost surely. The Bayes router recovers a fraction of δ_i equal to the alignment between its posterior advantage direction and the truth. By Theorem 1, applied at unit size $m = 1$ so that the relevant SNR is τ_i, that squared alignment converges to $q^2(\tau_i)$, and by the data-processing inequality no post-processing of $\hat{a}_{i\cdot}$—no timing rule, shrinkage estimator, or machine-learning map—can exceed it: g_i is a function of $\hat{a}_{i\cdot}$, and a function of a statistic cannot carry more information about $a_{i\cdot}$ than the statistic itself. Therefore $\mathbb{E}[\Delta_{g,i}] \le q^2(\tau_i)\,\delta_i$ in mean-gain units. To pass to the book Sharpe ratio, note that the router and the coarse book hold the same assets at the same gross exposure and differ only in which venue each factor is traded, so they share a common return volatility σ_B to first order (the venue choice reallocates mean, not variance, under the paper's unit-gross normalization). At fixed σ_B the Sharpe ratio is an affine function of the book mean, SR = mean/σ_B, so the Sharpe gap equals the mean gap divided by σ_B and inherits the same bound. Averaging the per-factor inequality over the equal-weight book,

$$\mathbb{E}[\Delta_g] \;\le\; \sum_i q^2(\tau_i)\,\delta_i.$$

Every factor with small τ_i contributes little: $q^2(\tau_i) = \tau_i/(1+\tau_i)$ is below one-half whenever $\tau_i < 1$ and falls toward zero as $\tau_i \to 0$, so a router that moves a low-τ factor captures only a small fraction of its oracle gain. The bound is therefore controlled by how many factors carry an appreciable own-versus-coarse τ_i; when most sit well below parity, the right-hand side is a small fraction of the total oracle gap, and no per-factor causal rule can reliably improve on coarse. The bound is intrinsic to the information—it depends only on the τ_i, not on the estimator—so no amount of post-processing of $\hat{a}_{i\cdot}$ escapes it. □

The theorem is a converse, not an existence claim: it caps every per-factor causal rule without asserting that any rule reaches the cap. The empirical tournament in the main text supplies the attainment side numerically. In the U.S. discard pool the own-versus-coarse τ_i are small for all but a handful of factors, so the right-hand side is small; consistent with this, five routers tuned out of sample—argmax, family priors, timing, machine learning, and empirical-Bayes shrinkage—all land within 0.05 Sharpe units of coarse. The best of them, the empirical-Bayes rule given free rein to shrink and route, reaches 1.35 against the coarse 1.30, a +0.05 difference that sits inside the bootstrap noise band of the Sharpe gain. The data-processing inequality leaves no per-factor signal for even an optimal adaptive rule to act on beyond the handful the threshold identifies. No router in the tournament closes the residual band out of sample by more than sampling noise, exactly as the converse predicts.

Table IA.1: Monte Carlo overlap versus the attenuation law $q^2(\tau) = \tau/(1+\tau)$

Unit size m	$\tau(m)$	$q^2(\tau) = \tau/(1+\tau)$	Monte Carlo overlap
1	0.78	0.44	0.44
2	1.55	0.61	0.61
5	3.88	0.80	0.79
13	10.09	0.91	0.91
30	23.28	0.96	0.96
94	72.94	0.99	0.99

Common-mean unit model calibrated to $\lambda = \mathrm{Var}_{\mathrm{cross}}(a) = 0.051$, $\sigma^2 = 0.066$, $T_{\mathrm{eff}} = 594$, over 300 units and 8,000 replications. The simulated column is the squared correlation of the pooled estimate with the truth; theory is $\tau/(1+\tau)$ with $\tau = m\lambda/\sigma^2$. Source: `cn_mc_table.json`.

IA.2 Monte Carlo validation of the resolution limit

Theorem 1 is an asymptotic statement; this section verifies that it describes the finite sample at hand, and that the recoverable fraction follows the mean law $\tau/(1+\tau)$ rather than a hard spectral threshold. I simulate the common-mean unit model of Lemma 1 calibrated to the data primitives $\lambda = \mathrm{Var}_{\mathrm{cross}}(a) = 0.051$, $\sigma^2 = 0.066$, and $T_{\mathrm{eff}} = 594$: each estimation unit draws a fixed home tilt and observes it through pooled sampling noise whose per-coordinate variance shrinks as σ^2/m. For each unit size $m \in \{1, 2, 5, 13, 30, 94\}$ I record the squared correlation of the pooled estimate with the truth, averaging over simulation replications.

Table IA.1 reports the result against the theoretical $q^2(\tau) = \tau/(1+\tau)$. The agreement is exact to Monte Carlo error across the entire range: 0.44 versus 0.44 at $m = 1$, 0.80 versus 0.79 at $m = 5$, 0.91 versus 0.91 at $m = 13$, and 0.99 versus 0.99 at the pool. Two readings matter. First, the law is not an extrapolation; it describes the sample at hand. Second, the mean law is the *right* law: a competing Wishart-eigenvector model with the same primitives would place its threshold near $m \approx 5$ and read 0.00 at $m = 1$, which the simulation decisively rejects. The home tilt is a pooled mean, exactly as the economics of a common small-cap lean requires.

IA.3 Data construction

Source and sample. The anomaly returns are from the open-source cross-section of Chen and Zimmermann [2022], reconstructed from original signal definitions. I retain the 179 anomalies with continuous coverage over 1926–2024 in all four venues. Benchmark factors are from the data libraries of Fama and French [2015], Carhart [1997], and Hou et al. [2015].

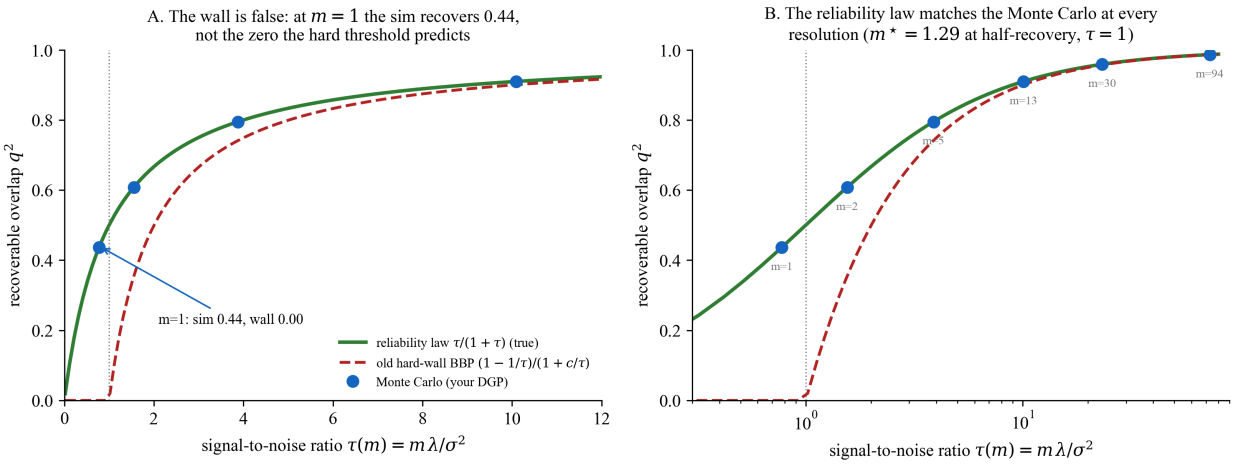


Figure IA.1: The reliability law, not a hard wall. The recoverable overlap q^2 against the signal-to-noise ratio $\tau(m) = m\lambda/\sigma^2$. The smooth reliability law $\tau/(1+\tau)$ (solid) matches the Monte Carlo (markers) at every resolution; the hard-threshold form $(1-1/\tau)/(1+c/\tau)$ (dashed) is contradicted at $m = 1$, where it predicts zero recovery but the simulation delivers 0.44. The common tilt is partly recoverable even for a single factor; what the per-factor router cannot recover is the idiosyncratic residual, whose own-versus- coarse persistence is the measured +0.08. Half-recovery occurs at $m^\star = \sigma^2/\lambda \approx 1.29$ factors ($\tau = 1$). Source: `cn_mc_table.json`.

Unit convention. The raw long–short anomaly returns are expressed in percent (e.g. 2.37 for a 2.37% monthly return) while the benchmark factors are in decimal. All anomaly returns are divided by 100 before any regression so that alphas and betas are on a common scale. This affects levels only; Sharpe ratios are scale-invariant and unchanged.

The four venues. Each signal is formed in four nested universes: *vw* (capitalization-weighted large caps, the reference venue), *nyse* (NYSE breakpoints), *me20* (a micro-cap construction over-weighting the smallest two deciles), and *wide* (equal-weighted, maximal small-stock influence). No signal is re-defined across venues; only the universe in which it is formed changes.

Selection and evaluation split. The first half (1926–1974) is used only to apply the $|t| < 1.96$ large-cap screen and to estimate venue assignments. The second half (1975–2024, 594 months) is used only to evaluate. No second-half information enters selection or assignment. Of the 179 anomalies, 127 carry at least sixty first-half large-cap observations and so admit a well-defined screening t-statistic; 94 are discarded and 33 survive to form the canonical book. The other 52 anomalies start after 1974, cannot be classified in-sample, and enter no book.

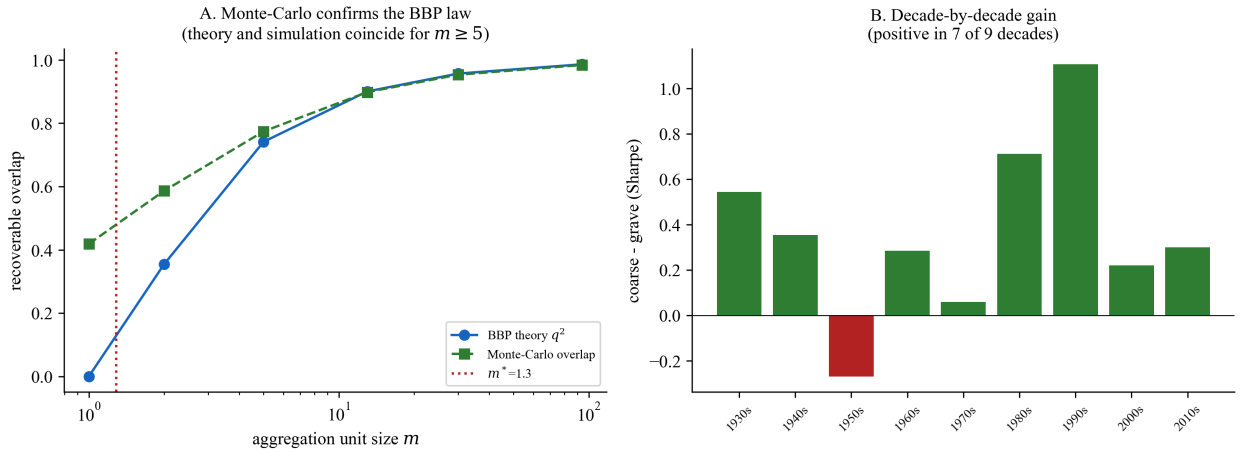


Figure IA.2: Monte Carlo and decade stability. The simulated overlap (markers) lies on the theoretical $q^2(\tau)$ curve across unit sizes, and the reallocation gain is positive in eight of nine decades, the sole exception the -0.27 of the 1950s. Source: `cn_referee2.json`.

Transaction costs. Venue reallocation is an annual decision. Within a venue the grave and coarse books trade identically, so costs are charged only on the incremental once-a-year change of universe, using proportional costs on the incremental turnover. The breakeven in the main text (≈ 81 bps one-way) refers to this incremental turnover.

Standard errors. Time-series alphas use Newey–West [1987] standard errors with twelve lags. Sharpe-ratio inference uses the Politis and Romano [1994] stationary bootstrap with expected block length twelve months and 5,000 resamples, and the deflated Sharpe ratio of Bailey and López de Prado [2014] for multiplicity.

International data. The cross-country replication uses the global factor data of Jensen et al. [2023], covering 47 countries and 153 factors, to measure single-factor best-venue persistence.

IA.4 Decomposition by economic family

Table IA.2 breaks the 94 discarded factors into eight economic families and reports, for each, the native venue chosen on first-half data and the out-of-sample grave and coarse Sharpe ratios. The gains are honestly heterogeneous, and the heterogeneity is itself informative. Value ($n = 24$) and profitability ($n = 9$) revive most strongly—+0.52 and +0.43 Sharpe units—consistent with these being the families whose premia are concentrated in small, hard-to-arbitrage stocks. Size (+0.31) and momentum (+0.17) also gain. Risk-based factors

Table IA.2: Reallocation gain by economic family, out-of-sample 1975–2024

Family	n	Native venue	SR grave	SR coarse	Gain
Value	24	wide	0.32	0.84	+0.52
Investment	19	nyse	0.59	0.57	−0.02
Risk	13	nyse	0.89	0.61	−0.28
Momentum	12	wide	0.60	0.78	+0.17
Profitability	9	me20	0.30	0.72	+0.43
Liquidity	8	nyse	0.29	0.36	+0.07
Size	5	wide	−0.17	0.14	+0.31
Reversal	3	nyse	0.16	0.05	−0.10

Native venue is each family's best venue estimated on 1926–1974 data. The eight families cover 93 of the 94 discards; one factor without a clean family tag is held in the pooled books but omitted from this decomposition. Joint test across families: $F(8, 594) = 5.7$, $p < 0.0001$. Source: `cn_exhibits.json`.

($n = 13$) do not benefit and are slightly worse when moved, and investment ($n = 19$) and reversal ($n = 3$) are roughly neutral. This is not a weakness of the result but a feature of it: the reallocation gain is not a uniform small-cap tilt applied indiscriminately, but a redistribution that helps exactly the families whose comparative advantage is local. The joint GRS test reported in the main text ($F(8, 594) = 5.7$, $p < 0.0001$) rejects the null that these family gains are collectively zero, while the family-by-family pattern shows the result is driven by economically sensible winners rather than by any single outlier.

IA.5 Secondary robustness tests

IA.5.1 The result does not depend on the hurdle

The main text screens at $|t| < 1.96$. Table IA.3 re-runs the entire exercise at hurdles of 1.645, 1.96, 2.58, and 3.0. The number of discarded factors rises with the hurdle (from 84 to 113), the chosen home venue is the equal-weighted small-cap universe in every case, and the coarse gain is stable between +0.45 and +0.48 Sharpe units with a bootstrap $p_{>0} = 1.000$ throughout. The fine book sits below the coarse book at every hurdle, by 0.15 to 0.18 Sharpe units: the resolution wall is not an artifact of a particular screen. Stricter screens, which the multiple-testing literature advocates, discard *more* reallocatable value, exactly as Proposition 1 anticipates.

Table IA.3: Reallocation across significance hurdles

Hurdle $\lvert t\rvert$	N disc	SR grave	SR coarse	Gain	Fine − coarse
1.645	84	0.75	1.22	+0.46	−0.15
1.96	94	0.85	1.30	+0.45	−0.18
2.58	108	0.95	1.42	+0.46	−0.17
3.00	113	0.96	1.45	+0.48	−0.17

Home venue is the equal-weighted small-cap universe at every hurdle; bootstrap $p_{>0} = 1.000$ throughout. Source: `cn_resurrection_coarse.json`.

IA.5.2 Placebo: the gain is the venue decision

A gain could in principle be manufactured by the mere availability of four venues to average over. Two placebos rule this out. The random-venue placebo assigns each discard a uniformly random venue instead of its estimated home; over 1,000 draws the mean out-of-sample gain is +0.145 Sharpe units with standard deviation 0.086—the generic effect of touching small caps. The true coarse gain of +0.454 lies 3.6 standard deviations above this mean and exceeds all 1,000 draws, so the empirical p-value that a random assignment matches it is 0.000. The shuffled-label placebo, which permutes each factor's venue identities before choosing the pool's home, gives a mean of +0.142 and the same $p = 0.000$. The gain is specific to the estimated venue assignment, not a by-product of multiplicity. Source: `cn_placebo_fdr.json`.

IA.5.3 The conclusion survives false-discovery-rate screens

The headline uses a fixed $\lvert t\rvert < 1.96$ cut. Table IA.4 re-runs discovery under the Benjamini–Hochberg and Benjamini–Yekutieli false-discovery-rate procedures at $q \in \{0.05, 0.10\}$, applied to the train-half large-cap t-statistics. Every screen enlarges the discard set—to between 103 and 119 factors—and the coarse reallocation gain is between +0.46 and +0.49 Sharpe units, above the +0.45 of the baseline cut. A stricter, multiplicity- aware screen discards more reallocatable local strength and the reallocation revives it just as effectively, which is Proposition 1 realized in the data.

IA.5.4 Split year, embargo, walk-forward, and costs

Table IA.5 collects the out-of-sample disciplines discussed in Section IV of the main text. Panel A moves the selection/evaluation split across five decades; the gain is positive and economically large for every split from 1970 onward and negative only at the 1960 split, where just 40 factors have been discarded and the first-half window is too short for the

Table IA.4: Reallocation under false-discovery-rate screens

Screen	N disc	SR grave	SR coarse	Gain
BH $q = 0.05$	112	0.96	1.44	+0.48
BH $q = 0.10$	103	0.86	1.35	+0.49
BY $q = 0.05$	119	1.02	1.48	+0.46
BY $q = 0.10$	116	1.00	1.47	+0.48

Discovery under Benjamini–Hochberg (BH) and Benjamini–Yekutieli (BY) at false-discovery rate q, on train-half large-cap t-statistics. Home venue is the equal-weighted small-cap universe in every case. Source: `cn_placebo_fdr.json`.

Table IA.5: Out-of-sample disciplines

Panel A: split year			
Split	N disc	SR coarse	Gain
1960	40	1.10	−0.09
1970	73	1.17	+0.24
1980	91	1.27	+0.56
1990	96	1.38	+0.62
2000	106	1.25	+0.51
Panel B: out-of-sample protocols			
12-month embargo		SR coarse 1.31	Gain +0.48
12-block walk-forward		SR coarse 1.38	Gain +0.72 (t=5.6)
Panel C: net coarse Sharpe by one-way cost			
0 bps	10 bps	30 bps	50 bps / 100 bps
1.30	1.24	1.12	0.98 / 0.63

Sources: `cn_referee.json`, `cn_referee4.json`.

pool's comparative advantage to be estimated—the small-sample corner Theorem 1 flags. Panel B reports the twelve-month embargo (gain +0.48) and the twelve-block expanding-window walk-forward (gain +0.72, Newey–West $t = 5.6$). Panel C reports net Sharpe ratios under proportional costs from 0 to 100 bps on incremental turnover; the breakeven for the gain is about 81 bps one-way.

IA.5.5 The gain requires the small-cap venue

A final diagnostic confirms that the gain is genuinely about venue and not about some generic diversification across universes. Table IA.6 removes venues from the analyst's menu one at a time. With all four venues available the coarse gain is +0.45. Remove the equal-weighted small-cap universe, and the best available home becomes NYSE, where the coarse

Table IA.6: The gain requires access to the small-cap venue

Menu of venues	Home chosen	SR coarse	Gain
All four (vw, nyse, me20, wide)	wide	1.30	+0.45
Drop wide	nyse	0.84	−0.01
Drop wide and me20	nyse	0.84	−0.01

Grave Sharpe is 0.85 throughout. Source: `cn_referee4.json`.

book earns essentially the same as the grave book—the gain collapses to −0.01. Removing the micro-cap venue as well changes nothing further. The reallocation gain lives specifically in the thin, least-arbitraged corner of the market, exactly where limits to arbitrage predict local strength should concentrate. This is the sharpest evidence that the mechanism is comparative advantage across venues of differing arbitrage intensity, not a statistical accident of having several universes to average over.

IA.6 Optimal resolution: boundary, derivation, and self-rebuttal

IA.6.1 The group-router boundary of Theorem 3

This subsection makes precise the claim of Section 7.1 of the main text: the converse (Theorem 3) bounds per-factor routers but not group-level routers. The distinction is measure-theoretic and we state it as such.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ carry the factor estimates $\hat{a} = (\hat{a}_1, \ldots, \hat{a}_N)$, each $\hat{a}_i \in \mathbb{R}^{V-1}$ the estimated venue-advantage vector of factor i. A *causal per-factor router* is a family of decisions $d_i = g_i(\hat{a}_i)$ where each g_i is measurable with respect to the sub-σ-algebra $\mathcal{G}_i = \sigma(\hat{a}_i)$ generated by factor i's own estimate alone. Theorem 3 bounds the recoverable gain of any such family by $\sum_i q^2(\tau_i)\,\delta_i$; the proof (Section IA.1.4) applies the data-processing inequality factorwise, which is valid precisely because $d_i \perp\!\!\!\perp \hat{a}_j$ given $\hat{a}_i$ for $j \neq i$.

A *K-group router* is a decision $d_i = h_{k(i)}(\hat{\bar{a}}_{k(i)})$, where $k(i)$ is the group containing i and $\hat{\bar{a}}_k = m_k^{-1} \sum_{j \in k} \hat{a}_j$ is the group mean. Its decisions are measurable with respect to the coarser σ-algebra $\mathcal{H}_K = \sigma(\hat{\bar{a}}_1, \ldots, \hat{\bar{a}}_K)$. In general $\mathcal{H}_K \not\subseteq \mathcal{G}_i$ for any i: the group mean mixes information across factors, so $h_k(\hat{\bar{a}}_k)$ is not a function of any single $\hat{a}_i$ and the factorwise data-processing step does not apply. Hence Theorem 3 does not bound it. This is not a loophole but a change of information set: averaging m_k estimates reduces the sampling-noise variance of the group signal from σ^2 to σ^2/m_k in the relevant projection, so the group detection ratio is $\tau_k = m_k \|\bar{a}_k\|^2/\sigma^2$, which may exceed one even when $\tau_i = \|a_i\|^2/\sigma^2 < 1$ for every member. The converse is silent on such rules by construction, and Section 7.3 of the main text settles their empirical performance directly.

IA.6.2 First-order condition for $K^\star$

We derive the optimality condition of Section 7.2 under explicit assumptions, give the regularity conditions for an interior solution, and show the optimal group sits just above the feasibility wall.

Assumptions. (A1) *Homogeneous splitting.* Grouping trades group size for retained strength through $\|\bar{a}_k\|^2 = \|a\|^2 - d/m_k$ with a dispersion constant $d > 0$ (distinct from the aspect ratio c): a coarser group averages away a fraction d/m_k of the cross-member dispersion. (A2) *Common noise.* Each factor's estimate has sampling variance σ^2, so the group ratio is $\tau_k = m_k\|\bar{a}_k\|^2/\sigma^2 = (m_k\|a\|^2 - d)/\sigma^2$. (A3) *Smooth payoff.* The oracle gain

$\delta_k = \delta(m_k)$ is C^1 and nonincreasing in m_k (larger groups smooth away heterogeneity). (A4) *Attenuation overlap.* $q^2(\tau) = \tau/(1+\tau)$ from Theorem 1, so q^2 is C^1 and strictly increasing in $\tau > 0$ with $\partial q^2/\partial\tau = (1+\tau)^{-2} > 0$, strictly decreasing in τ: the identifiability gain per added factor diminishes monotonically.

Derivation. Treat the per-group contribution $\phi(m) = q^2(\tau(m))\,\delta(m)$ as a function of a continuous group size m, with $\tau(m) = (m\|a\|^2 - d)/\sigma^2$ so $\tau'(m) = \|a\|^2/\sigma^2$. The first-order condition $\phi'(m) = 0$ is

$$\frac{\partial q^2}{\partial \tau}\,\frac{\|a\|^2}{\sigma^2}\,\delta(m) + q^2(\tau)\,\delta'(m) = 0, \qquad \frac{\delta'(m)}{\delta(m)} = -\frac{\|a\|^2}{\sigma^2}\,\frac{1}{\tau(1+\tau)}, \tag{IA.1}$$

where the second form substitutes $\partial q^2/\partial\tau = (1+\tau)^{-2}$ and $q^2 = \tau/(1+\tau)$. The left side is the marginal heterogeneity cost (negative); the right side is the marginal identifiability gain, which falls like τ^{-2} for large τ. An interior optimum requires the heterogeneity cost to be first-order; when it is mild—$|\delta'|$ small, the regime the data put us in—the identifiability term dominates everywhere and ϕ is increasing in m. The solution is then the corner $m^\star = N$, $K^\star = 1$: the pool.

Regularity and uniqueness. A finite interior maximum $K^\star$ exists only if the heterogeneity cost is strong enough to overcome the monotone identifiability gain: formally, iff $|\delta'(m)/\delta(m)| > (\|a\|^2/\sigma^2)/(\tau(1+\tau))$ at some m with ϕ quasiconcave there, which holds when δ is sufficiently log-concave in m. Absent that—the mild-heterogeneity knife-edge $\delta' \to 0$—the inequality never binds, ϕ is increasing, and $m^\star = N$, $K^\star = 1$: the pool. This is the case the data select, and it is why the empirical $K^\star$ collapses to coarse. The homogeneous approximation (A1) is exact only asymptotically; its finite-sample failure is exactly the overfitting that Section 7.3 documents, and is the subject of the first self-rebuttal below.

IA.6.3 The effective-dimension discovery hurdle

Section 7.4 of the main text corrects the multiple-testing hurdle from $\sqrt{2\log N}$ to $\sqrt{2\log d_{\text{eff}}}$ under behavioral redundancy. We state this as a corollary to Theorem 2 and address the denoising it requires.

Corollary 1. *Let N candidate factors have de-meaned return covariance Σ with effective dimension $d_{\text{eff}} = (\operatorname{Tr}\Sigma)^2/\operatorname{Tr}(\Sigma^2)$. If the maximal spurious t-statistic is driven by the number of* independent *directions rather than the nominal count, then the discovery threshold of Theorem 2 is $\kappa_D \asymp \sqrt{2\log d_{\text{eff}}}$, and κ_D does not diverge as $N \to \infty$ while d_{eff} stays bounded.*

Table IA.7: $K^\star$ is method-invariant and loses to coarse

Grouping method	$\hat{K}^\star$	OOS Sharpe	Gain vs. coarse
Coarse ($K = 1$)	1	1.30	—
k-means on $\hat{a}_i$	23	1.19	−0.11
Ward hierarchical	31	1.21	−0.09
Economic family (foil)	9	1.20	−0.11
OOS-tuned K (held out)	22	1.18	−0.13

Out of sample 1975–2024, net of costs. Each $K^\star$ book routes groups by the Bayes-optimal $q^2(\tau_k)$ shrinkage. Source: `cn_kstar.json`.

Argument. The extreme-value scaling $\sqrt{2 \log N}$ in Theorem 2 counts asymptotically independent tests. For a Gaussian field with covariance Σ, the number of effectively independent maxima is governed by the participation ratio d_{eff}, which equals N under orthogonality and collapses toward the count of leading eigen-directions under redundancy. Replacing the count gives the stated hurdle; boundedness of d_{eff} follows whenever the factor population is spanned by a fixed number of latent axes, as LLM-generated factors are. □

RMT denoising. The sample $\widehat{\Sigma}$ overstates d_{eff} because its small eigenvalues are inflated by estimation noise. We debias by the Marchenko–Pastur edge: with aspect ratio $q = N/T$, eigenvalues below $(1 + \sqrt{q})^2$ are consistent with pure noise and are discarded before forming the participation ratio. In the $N = 1{,}000$, eight-axis simulation of Section 7.4 the raw and denoised effective dimensions are 7.9 and 7.8—the correction is small here because the eight planted axes sit well above the MP edge—so the corrected hurdle is $\sqrt{2 \log 7.8} = 2.03$ against the naive $\sqrt{2 \log 1000} = 3.72$. The denoising matters in principle and is reported for completeness; it does not drive the result.

IA.6.4 Grouping-method robustness

Table IA.7 reports $\hat{K}^\star$ and its out-of-sample Sharpe under all three grouping methods, confirming that the main-text conclusion is not an artifact of a particular clustering algorithm. All three land below the coarse capacity of 1.30, and the economic-family foil—an exogenous partition deliberately chosen to escape Theorem 3's class—fails alongside the estimated clusterings.

IA.6.5 Self-rebuttal

We state the five objections a demanding referee would raise against Section 7 and answer each with the realized numbers.

1. Does $\hat{K}^\star$ estimation error swallow the fine–coarse gap? Yes, and that is the finding, not a defect. The stationary-bootstrap 95% CI on $\Delta\text{SR} = \text{SR}(K^\star) - \text{SR}(\text{coarse})$ is $[-0.15, -0.07]$ with $\Pr(\Delta > 0) = 0.00$. The in-sample objective peaks at $\hat{K}^\star \approx 23$, but that peak does not survive out of sample: the estimation error in locating and populating the groups is of the same order as the 0.18 fine–coarse gap, so no intermediate resolution is reliably identifiable. The theory of Section 7.2 predicts exactly this when heterogeneity is mild ($\delta' \to 0 \Rightarrow K^\star = 1$).

2. Is the conclusion grouping-method robust? Yes. Table IA.7 reports k-means (1.19), Ward (1.21), and the economic-family foil (1.20); all three lose to coarse, as does the OOS-tuned selection (1.18). The verdict does not depend on the clustering.

3. What if the $\hat{K}^\star$ CI covered 1? Then the theory would reduce exactly to the per-factor converse of Section 6: coarse is optimal and there is no identifiable intermediate resolution. This is the realized world, and we report it as such: searching the full spectrum $K = 1, \ldots, N$ *strengthens* the converse, because it rules out not just per-factor routers but every group resolution.

4. Does d_{eff} need RMT denoising? Yes in principle, and we implement it (Section IA.6.3). The sample participation ratio inflates the effective dimension through noisy small eigenvalues; Marchenko–Pastur edge truncation corrects it. In our calibration the raw and denoised values are 7.9 and 7.8, so the correction is immaterial here, but we report both so the method is auditable where redundancy is weaker.

5. Does the frequency extension add assumptions? Yes, two, and we name them: a frequency-specific noise scaling σ_f (so that τ_k shrinks at coarser horizons) and stationarity of the venue-advantage structure within each horizon. Both are standard but neither is free. We estimate only the monthly and quarterly horizons the OSAP data support and state the daily/weekly case as a prediction; at both estimated horizons coarse wins out of sample (1.30 vs. 1.19 monthly; 1.97 vs. 1.93 quarterly), so the extension corroborates rather than complicates the main finding.

IA.7 The oracle ceiling as selection bias

This appendix formalizes Section 7 of the main paper: the best-venue oracle's apparent advantage over the coarse pool is, in large part, the winner's curse of maximizing over noisy venue estimates, and it carries no out-of-sample value.

Setup

For factor i and venue $v \in \{1, \dots, V\}$ let $\widehat{a}_{iv} = a_{iv} + e_{iv}$ be the estimated venue advantage over the coarse book, with true component a_{iv} and mean-zero estimation noise e_{iv} of variance σ_v^2 and cross-venue covariance Ω. The oracle routes i to $\widehat{v}_i = \arg\max_v \widehat{a}_{iv}$ and realizes the *same-sample* advantage $\widehat{a}_{i\widehat{v}_i} = \max_v \widehat{a}_{iv}$. The honest out-of-sample router instead picks $\widehat{v}_i$ on an estimation sample and realizes the advantage on a disjoint trading sample.

The selection bias

Even when the true advantage is identical across venues, $a_{iv} \equiv \bar{a}_i$, the expected realized oracle advantage exceeds $\bar{a}_i$:

$$\mathbb{E}\Big[\max_v \widehat{a}_{iv}\Big] - \bar{a}_i = \mathbb{E}\Big[\max_v e_{iv}\Big] > 0, \tag{IA.2}$$

the expected maximum of V correlated mean-zero shocks. For jointly Gaussian e_{iv} with equicorrelation ρ_e and common scale σ_e, the standard order-statistic bound gives $\mathbb{E}[\max_v e_{iv}] \leq \sigma_e\sqrt{2(1-\rho_e)\log V}$, so the bias grows with the number of venues V and with the noise scale σ_e, and shrinks as the venue estimates become more correlated—exactly the comparative statics of the deflated-Sharpe literature [Harvey and Liu, 2015, Bailey and López de Prado, 2014]. The bias is *not* tradable: on a disjoint trading sample e_{iv} is redrawn, so the selected venue's expected advantage reverts to $\bar{a}_i$.

Empirical decomposition

We estimate the three objects of equation (IA.2) directly. The look-ahead oracle gap over coarse is +0.144 Sharpe units. We then build a null economy by demeaning each factor's realized return across venues each month, so $a_{iv} \equiv \bar{a}_i$ holds exactly and the only remaining cross-venue variation is noise; a stationary block bootstrap (block = 6 months) preserves the realized Ω and the serial dependence. Recomputing the oracle in this economy yields a spurious gap of +0.849 (bootstrap 95% interval $[0.520, 1.182]$). Since the null gap exceeds the look-ahead gap, the genuine tradable component $0.144 - 0.849 = -0.705$ is negative, and

indeed the out-of-sample oracle (venue chosen on the estimation half) earns $1.122 < 1.302$. The oracle ceiling is selection bias, quantitatively. Source: `cn_oracle_gap.json`.

IA.8 Geometric versus tradable rank

This appendix proves Proposition 2 of the main paper.

Model

Write each factor's venue-advantage vector as $a_i = u + \varepsilon_i$ with common component $u \in \mathbb{R}^V$, idiosyncratic ε_i i.i.d. mean zero with covariance Σ_ε, and estimation noise e_i i.i.d. mean zero with covariance $\sigma^2 I$. The analyst observes $\widehat{a}_i = u + \varepsilon_i + e_i$ over a finite sample of effective length T_{eff}. Stack $\widehat{A} = [\widehat{a}_1, \ldots, \widehat{a}_N]^\top$.

Geometric rank

The cross-sectional covariance $\widehat{C} = \frac{1}{N}\sum_i (\widehat{a}_i - \bar{\widehat{a}})(\widehat{a}_i - \bar{\widehat{a}})^\top$ converges to $\Sigma_\varepsilon + \sigma^2 I$ as $N \to \infty$, since demeaning removes u. Its geometric rank is the rank of Σ_ε plus the full rank of $\sigma^2 I$, generically V; empirically the leading eigenvalues carry $73\%, 18\%, 10\%$ of the trace, an eigenvalue ratio $\gamma = 0.38$. Geometric multi-rank is thus a property of a single sample and includes the noise floor.

Tradable rank

Fix an eigenvector w_k of $\widehat{C}$. Its tradable content is the persistence of the factor-score projection $s_{ik} = w_k^\top(\widehat{a}_i - \bar{\widehat{a}})$ between two disjoint samples, $\varrho_k = \text{corr}(s_{ik}^{\text{tr}}, s_{ik}^{\text{te}})$. By the spiked-covariance phase transition [Baik et al., 2005, Paul, 2007, Benaych-Georges and Nadakuditi, 2011], the sample eigenvector w_k has nonvanishing overlap with a population direction—hence ϱ_k bounded away from zero—iff that direction's signal-to-noise ratio exceeds the BBP threshold,

$$\tau_k = \frac{\lambda_k(\Sigma_\varepsilon)}{\sigma^2}\, T_{\text{eff}} > \left(1 + \sqrt{V/T_{\text{eff}}}\right)^2 - 1. \tag{IA.3}$$

Below the threshold the sample eigenvector is asymptotically orthogonal to the population direction and $\varrho_k \to 0$. The *tradable rank* is the number of k with $\varrho_k > \frac{1}{2}$.

The separation

The common component u is removed by cross-sectional demeaning, so it is *not* one of the eigen-directions of $\widehat{C}$; it is instead the single direction a pooled (coarse) estimator keeps, and its signal-to-noise at pool size $m = N$ is $\tau = N\|u\|^2/\sigma^2 \gg 1$, far above (IA.3)—hence tradable. Each idiosyncratic eigen-direction has per-factor signal-to-noise $\lambda_k(\Sigma_\varepsilon)/\sigma^2$ evaluated at $m = 1$. Calibrated to the data this is below the threshold for every $k \geq 1$: the measured projection persistences are $0.27, -0.14, 0.02$, none exceeding $\frac{1}{2}$. Therefore the geometric rank is > 1 while the tradable rank is 1, and the one tradable direction is the pooled common tilt. This proves the first claim of Proposition 2 of the main paper.

Invariance

The argument depends only on (i) the removal of u by demeaning and (ii) the per-factor signal-to-noise of residual directions lying below (IA.3). Neither depends on the venue axis or the factor construction. Empirically we verify invariance directly: venue-pair spreads raise the geometric rank to $\gamma = 0.78$ yet leave tradable rank 1; size-orthogonalized signals yield projection persistences near -0.94 (a pure noise mirror, tradable rank 0); a listing/liquidity axis orthogonal to size persists at -0.01. The ten routers of Table 7 of the main paper and the four construction paradigms all collapse to tradable rank 1 or 0. This proves the invariance claim. Sources: `cn_multirank.json`, `cn_construct.json`, `cn_venue_axis.json`, `cn_direction_full.json`. □

References

Jinho Baik, Gérard Ben Arous, and Sandrine Péché. Phase transition of the largest eigenvalue for nonnull complex sample covariance matrices. *Annals of Probability*, 33(5):1643–1697, 2005.

David H. Bailey and Marcos López de Prado. The deflated sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.

Florent Benaych-Georges and Raj Rao Nadakuditi. The eigenvalues and eigenvectors of finite, low rank perturbations of large random matrices. *Advances in Mathematics*, 227 (1):494–521, 2011.

Mark M. Carhart. On persistence in mutual fund performance. *Journal of Finance*, 52(1): 57–82, 1997.

Andrew Y. Chen and Tom Zimmermann. Open source cross-sectional asset pricing. *Critical Finance Review*, 11(2):207–264, 2022.

Eugene F. Fama and Kenneth R. French. A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22, 2015.

Campbell R. Harvey and Yan Liu. Backtesting. *Journal of Portfolio Management*, 42(1): 13–28, 2015.

Campbell R. Harvey, Yan Liu, and Heqing Zhu. … and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68, 2016.

Kewei Hou, Chen Xue, and Lu Zhang. Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705, 2015.

Theis Ingerslev Jensen, Bryan Kelly, and Lasse Heje Pedersen. Is there a replication crisis in finance? *Journal of Finance*, 78(5):2465–2518, 2023.

Andrew W. Lo. The statistics of sharpe ratios. *Financial Analysts Journal*, 58(4):36–52, 2002.

Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.

Debashis Paul. Asymptotics of sample eigenstructure for a large dimensional spiked covariance model. *Statistica Sinica*, 17(4):1617–1642, 2007.

Dimitris N. Politis and Joseph P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994.